

Machine Programming

Lecture 9 – Agentic Frameworks: Model Context Protocol (2)

Ziyang Li

Logistics – Week 5

- Assignment 2

Assignment 2 - Evaluating Coding LLMs

SEP 25, 2026 6:08 AM

OCT 9, 2026 11:59 PM

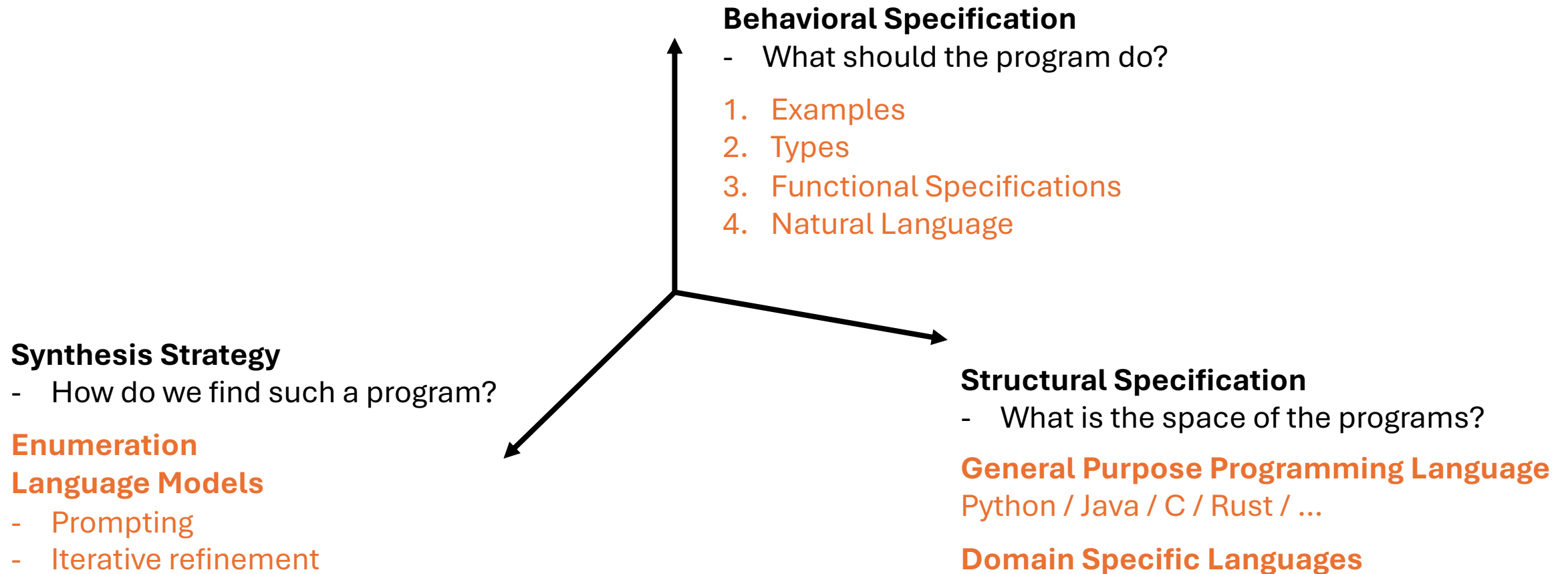
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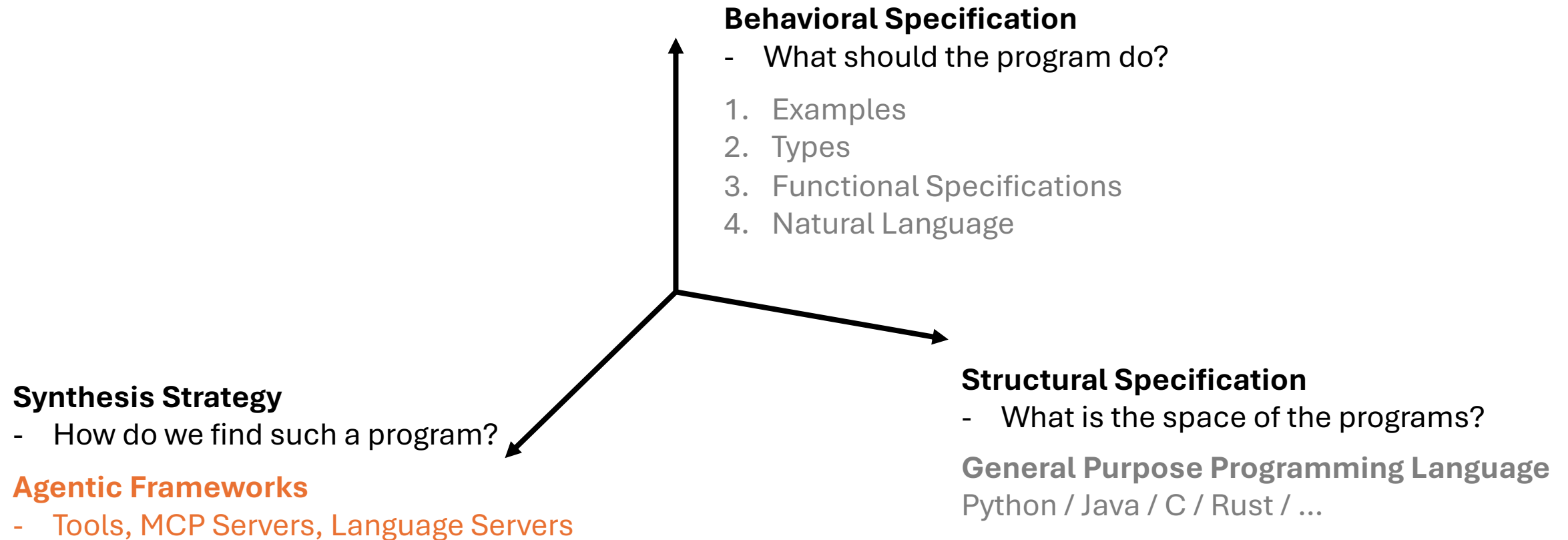
ON

- Due in two weeks (Oct 9) since release last Friday
- In case your LLM API key is running out of credit, please register for your own key following our tutorial!

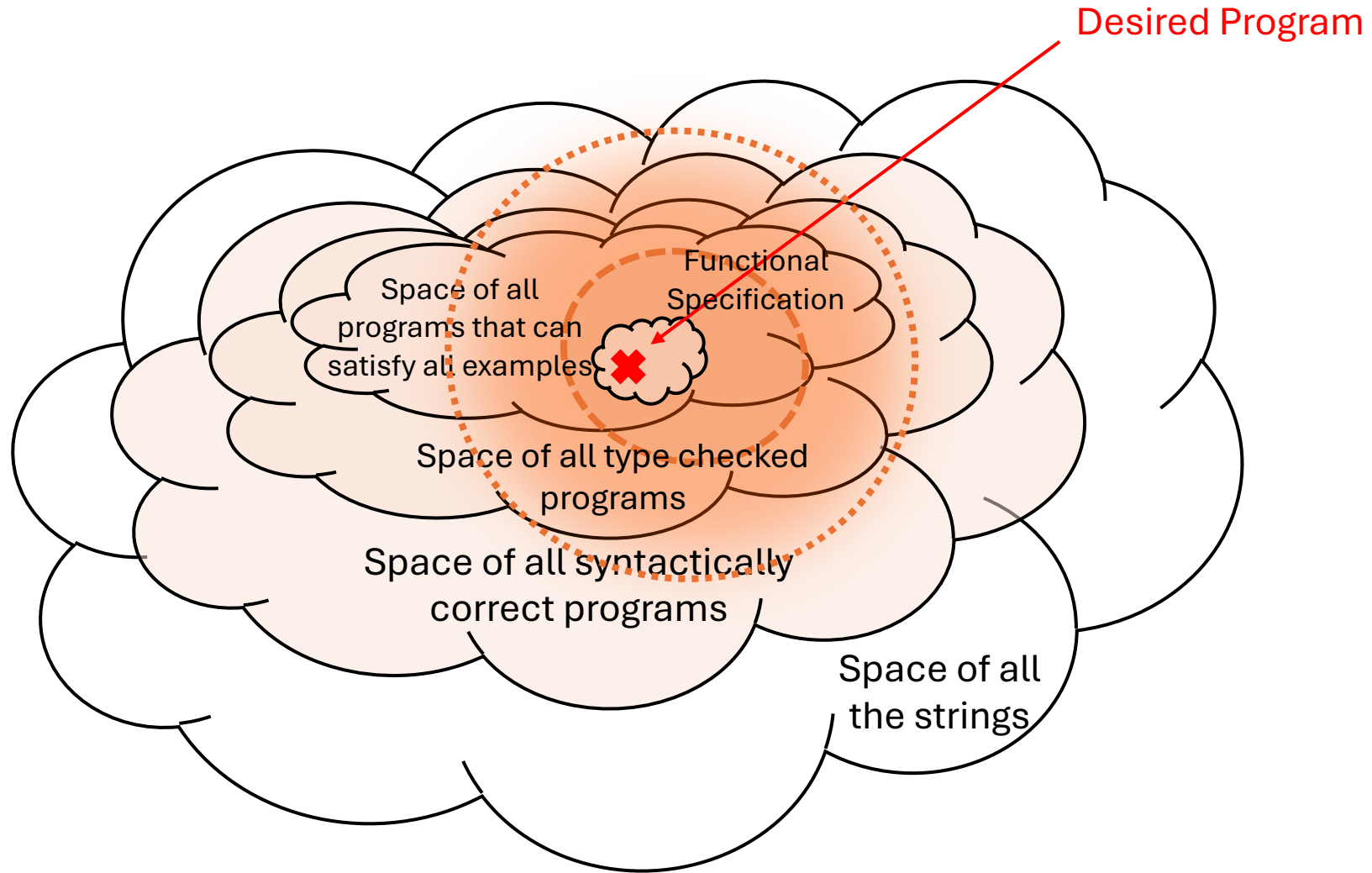
The Course So Far



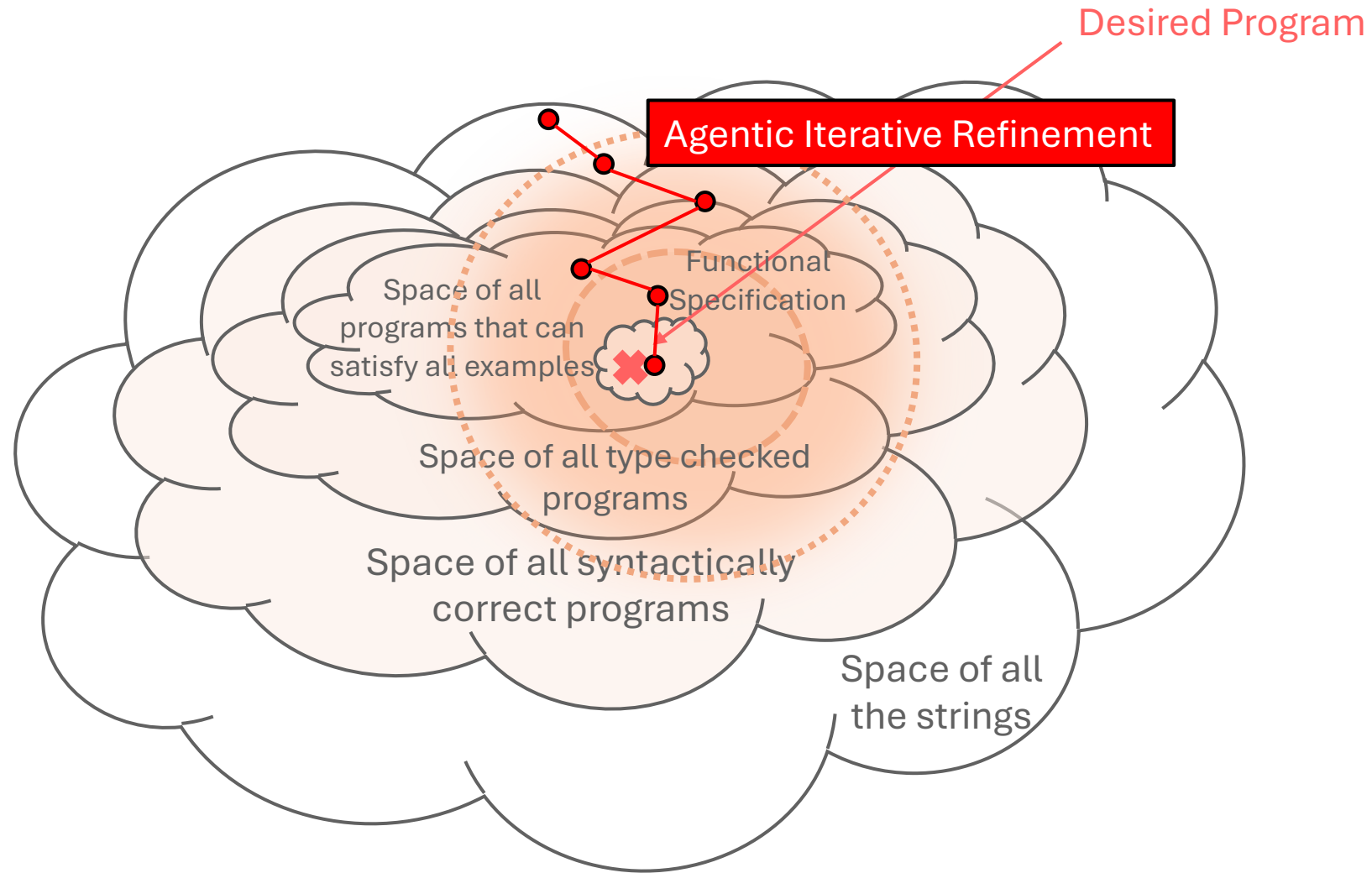
Today



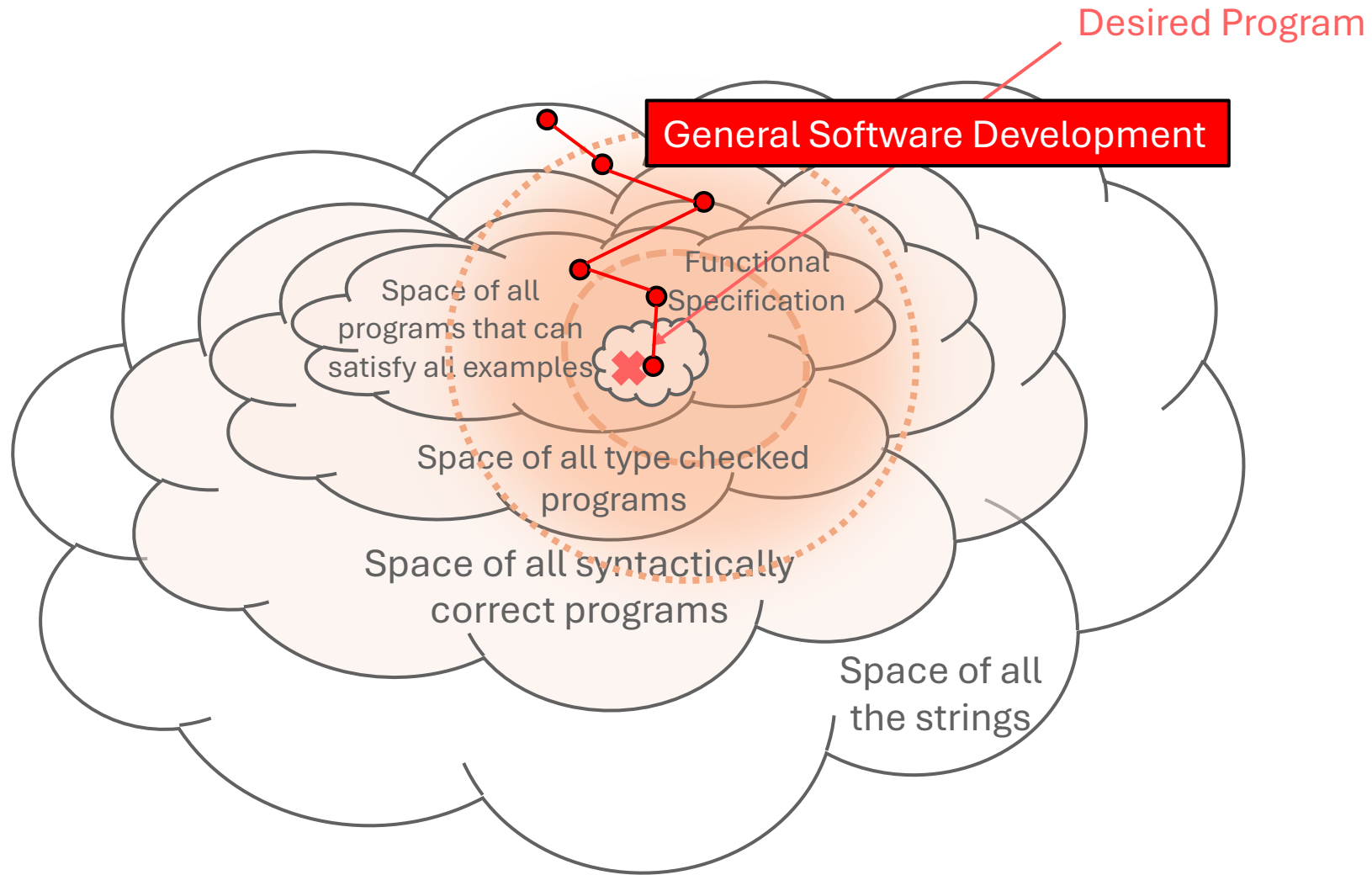
High Level Picture



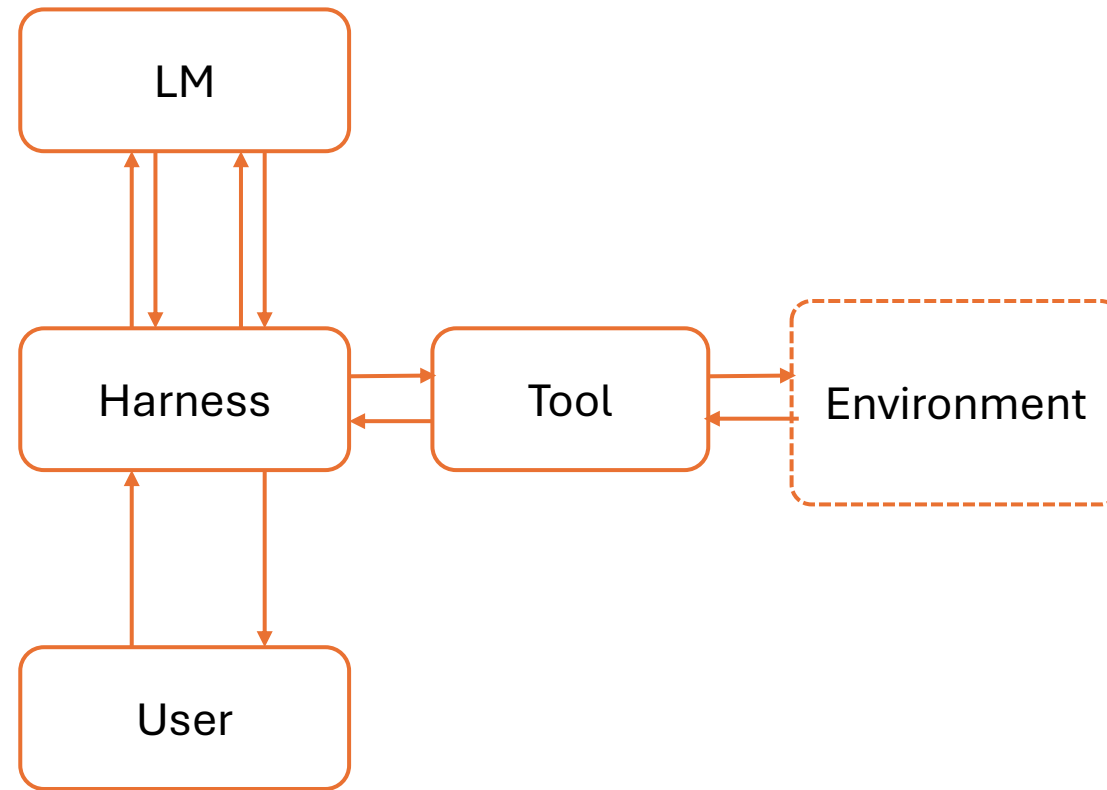
High Level Picture



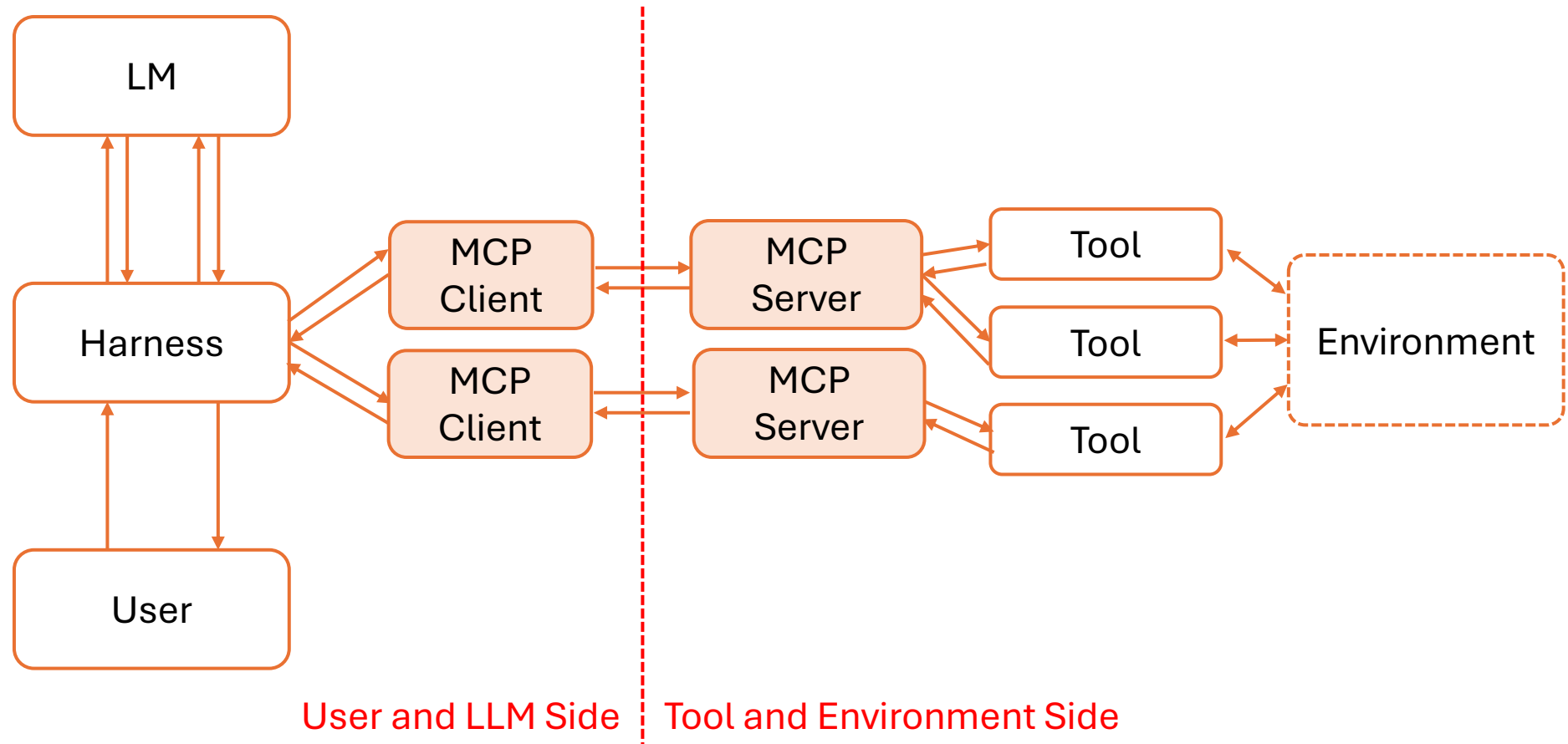
High Level Picture



Agentic Framework



Model Context Protocol (MCP)



Third-Party Servers

Official Integrations

Official integrations are maintained by companies building production ready MCP servers for their platforms.

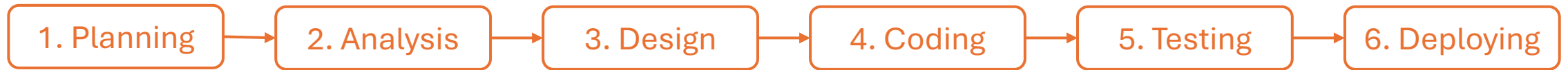
- 🔗 **21st.dev Magic** - Create crafted UI components inspired by the best 21st.dev design engineers.
- 🔗 **ActionKit by Paragon** - Connect to 130+ SaaS integrations (e.g. Slack, Salesforce, Gmail) with Paragon's **ActionKit** API.
- 🔗 **Adfin** - The only platform you need to get paid - all payments in one place, invoicing and accounting reconciliations with **Adfin**.
- 🔗 **AgentOps** - Provide observability and tracing for debugging AI agents with **AgentOps** API.
- 🔗 **AgentQL** - Enable AI agents to get structured data from unstructured web with **AgentQL**.
- 🔗 **AgentRPC** - Connect to any function, any language, across network boundaries using **AgentRPC**.
- 🔗 **Agentset** - RAG for your knowledge base connected to **Agentset**.
- 🔗 **Aiven** - Navigate your **Aiven projects** and interact with the PostgreSQL®, Apache Kafka®, ClickHouse® and OpenSearch® services
- 🔗 **Alation** - Unlock the power of the enterprise Data Catalog by harnessing tools provided by the Alation MCP server.
- 🔗 **Alby Bitcoin Payments** - Connect any bitcoin lightning wallet to your agent to send and receive instant payments globally with your agent.
- 🔗 **Algolia** - Use AI agents to provision, configure, and query your **Algolia** search indices.
- 🔗 **Alibaba Cloud AnalyticDB for MySQL** - Connect to an **AnalyticDB for MySQL** cluster for getting database or table metadata, querying and analyzing data. It will be supported to add the OpenAPI for cluster operation in the future.
- 🔗 **Alibaba Cloud AnalyticDB for PostgreSQL** - An MCP server to connect to **AnalyticDB for PostgreSQL** instances, query and analyze data.
- 🔗 **Alibaba Cloud DataWorks** - A Model Context Protocol (MCP) server that provides tools for AI, allowing it to interact with the **DataWorks** Open API through a standardized interface. This implementation is based on the Alibaba Cloud Open API and enables AI agents to perform cloud resources operations seamlessly.
- 🔗 **Alibaba Cloud OpenSearch** - This MCP server equips AI Agents with tools to interact with **OpenSearch** through a standardized and extensible interface.
- 🔗 **Alibaba Cloud OPS** - Manage the lifecycle of your Alibaba Cloud resources with **CloudOps Orchestration Service** and Alibaba Cloud OpenAPI.
- 🔗 **Alibaba Cloud RDS** - An MCP server designed to interact with the Alibaba Cloud RDS OpenAPI, enabling programmatic management of RDS resources via an LLM.
- 🔗 **AlipayPlus** - Connect your AI Agents to AlipayPlus Checkout Payment.
- 🔗 **AlVoiceLab** - An AI voice toolkit with TTS, voice cloning, and video translation, now available as an MCP server for smarter agent integration.
- 🔗 **Alpaca** - Alpaca's MCP server lets you trade stocks and options, analyze market data, and build strategies through **Alpaca's Trading API**
- 🔗 **AlphaVantage** - Connect to 100+ APIs for financial market data, including stock prices, fundamentals, and more from **AlphaVantage**
- 🔗 **AltTester®** - Use AltTester® capabilities to connect and test your Unity or Unreal game. Write game test automation faster and smarter, using **AltTester** and the AltTester® MCP server.
- 🔗 **Antom** - Connect your AI Agents to Antom Checkout Payment.
- 🔗 **Anypoint** - An MCP server enabling AI assistants to interact with **Anypoint** - a local and collaborative wiki - to organize objects, lists, and more through natural language.
- 🔗 **Apache Doris** - MCP Server For **Apache Doris**, an MPP-based real-time data warehouse.
- 🔗 **Apache IoTDB** - MCP Server for **Apache IoTDB** database and its tools
- 🔗 **Apache Pinot** - MCP server for running real - time analytics queries on Apache Pinot, an open-source OLAP database built for high-throughput, low-latency powering real-time applications.
- 🔗 **Apify** - Use 6,000+ pre-built cloud tools to extract data from websites, e-commerce, social media, search engines, maps, and more
- 🔗 **APIMatic MCP** - APIMatic MCP Server is used to validate OpenAPI specifications using **APIMatic**. The server processes OpenAPI files and returns validation summaries by leveraging APIMatic's API.
- 🔗 **Apollo MCP Server** - Connect your GraphQL APIs to AI agents
- 🔗 **Aqara MCP Server** - Control **Aqara** smart home devices, query status, execute scenes, and much more using natural language.
- 🔗 **Archbee** - Write and publish documentation that becomes the trusted source for instant answers with AI. Stop cobbling tools and use **Archbee** — the first complete

- 🔗 **Astra DB** - Comprehensive tools for managing collections and documents in a **AstraDB** NoSQL database with a full range of operations such as create, update, delete, find, and associated bulk actions.
- 🔗 **Atla** - Enable AI agents to interact with the **Atla API** for state-of-the-art LLMJ evaluation.
- 🔗 **Atlan** - The Atlan Model Context Protocol server allows you to interact with the **Atlan** servers through multiple tools.
- 🔗 **Atlassian** - Securely interact with Jira work items and Confluence pages, and search across both.
- 🔗 **AtomGit** - Official AtomGit server for integration with repository management, PRs, issues, branches, labels, and more.
- 🔗 **Audiense Insights** - Marketing insights and audience analysis from **Audiense** reports, covering demographic, cultural, influencer, and content engagement analysis.
- 🔗 **Auth0** - MCP server for interacting with your Auth0 tenant, supporting creating and modifying actions, applications, forms, logs, resource servers, and more.
- 🔗 **Authenticator App - 2FA** - A secure MCP (Model Context Protocol) server that enables AI agents to interact with the Authenticator App.
- 🔗 **AWS** - Specialized MCP servers that bring AWS best practices directly to your development workflow.
- 🔗 **Axiom** - Query and analyze your Axiom logs, traces, and all other event data in natural language
- 🔗 **Azure** - The Azure MCP Server gives MCP Clients access to key Azure services and tools like Azure Storage, Cosmos DB, the Azure CLI, and more.
- 🔗 **Azure DevOps** - Interact with Azure DevOps services like repositories, work items, builds, releases, test plans, and code search.
- 🔗 **BackdoCKET** - Search, Retrieve, and Update your **BackdoCKET** data. This currently includes Claims, Matters, Contacts, Tasks and Advanced Searches. To easily use the Remote Mcp Server utilize the following url: **https://ai.backdoCKET.com/mcp**
- 🔗 **Baidu Map - Baidu Map MCP Server** provides tools for AI agents to interact with Baidu Maps APIs, enabling location-based services and geospatial data analysis.
- 🔗 **Bankless Onchain** - Query Onchain data, like ERC20 tokens, transaction history, smart contract state.
- 🔗 **Baserow** - Query data from Baserow self-hosted or SaaS databases using MCP integration.
- 🔗 **BICScan** - Risk score / asset holdings of EVM blockchain address (EOA, CA, ENS) and even domain names.
- 🔗 **Bitrise** - Chat with your builds, CI, and **more**.
- 🔗 **Boikot** - Learn about the ethical and unethical actions of major companies with **boikot.xyz**.
- 🔗 **BoldSign** - Search, request, and manage e-signature contracts effortlessly with **BoldSign**.
- 🔗 **Boost.space** - An MCP server integrating with **Boost.space** for centralized, automated business data from 2000+ sources.
- 🔗 **Box** - Interact with the Intelligent Content Management platform through Box AI.
- 🔗 **BrightData** - Discover, extract, and interact with the web - one interface powering automated access across the public internet.
- 🔗 **Browserbase** - Automate browser interactions in the cloud (e.g. web navigation, data extraction, form filling, and more)
- 🔗 **BrowserStack** - Access BrowserStack's **Test Platform** to debug, write and fix tests, do accessibility testing and more.
- 🔗 **Buildkite** - Exposing Buildkite data (pipelines, builds, jobs, tests) to AI tooling and editors.
- 🔗 **Buildable** (TypeScript) - Official MCP server for Buildable AI-powered development platform. Enables AI assistants to manage tasks, track progress, get project context, and collaborate with humans on software projects.
- 🔗 **BuiltWith** - Identify the technology stack behind any website.
- 🔗 **Burp Suite** - MCP Server extension allowing AI clients to connect to **Burp Suite**
- 🔗 **Cal.com** - Connect to the Cal.com API to schedule and manage bookings and appointments.
- 🔗 **Campertunity** - Search campgrounds around the world on campertunity, check availability, and provide booking links.
- 🔗 **Canva** - Provide AI - powered development assistance for **Canva** apps and integrations.
- 🔗 **Carbon Voice** - MCP Server that connects AI Agents to **Carbon Voice**. Create, manage, and interact with voice messages, conversations, direct messages, folders, voice memos, AI actions and more in **Carbon Voice**.
- 🔗 **Cartesia** - Connect to the **Cartesia** voice platform to perform text-to-speech, voice cloning etc.
- 🔗 **Cashfree** - **Cashfree Payments** official MCP server.
- 🔗 **CB Insights** - Use the **CB Insights** MCP Server to connect to **ChatGPT**

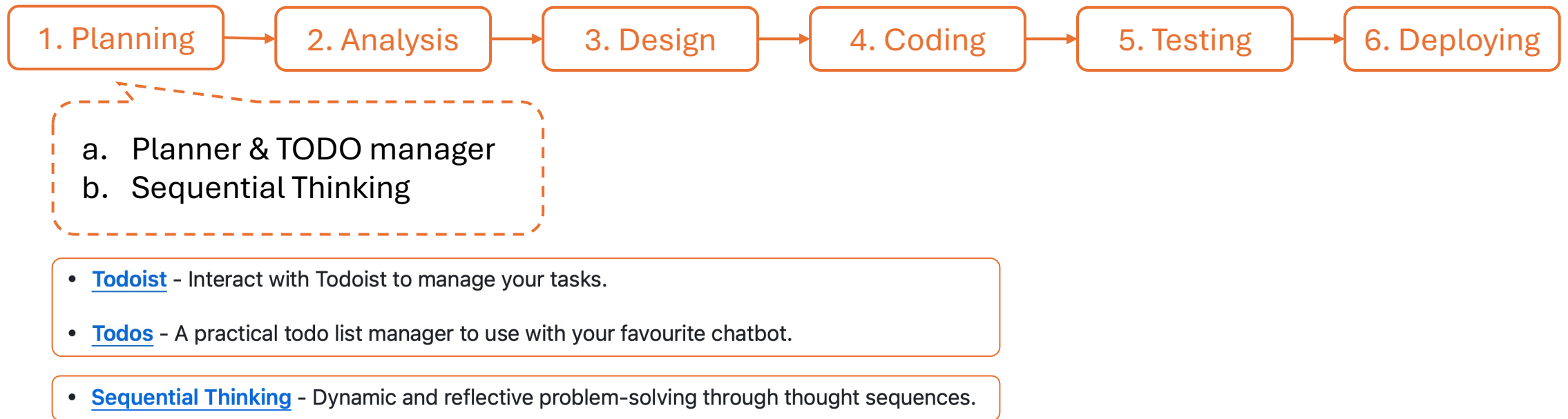
- 🔗 **Claudia** - Exposes **Claudia**'s media output, transformation, AI analysis, management, optimization and delivery as tools usable by AI agents
- 🔗 **Cloudsway SmartSearch** - Web search MCP server powered by Cloudsway, supporting keyword search, language, and safety options. Returns structured JSON results.
- 🔗 **Codacy** - Interact with **Codacy** API to query code quality issues, vulnerabilities, and coverage insights about your code.
- 🔗 **CodeLogic** - Interact with **CodeLogic**, a Software Intelligence platform that graphs complex code and data architecture dependencies, to boost AI accuracy and insight.
- 🔗 **CoinGecko** - Official **CoinGecko API** MCP Server for Crypto Price & Market Data, across 200+ Blockchain Networks and 8M+ Tokens.
- 🔗 **Comet Opik** - Query and analyze your **Opik** logs, traces, prompts and all other telemetry data from your LLMs in natural language.
- 🔗 **Conductor** - Interact with Conductor (OSS and Orkes) REST APIs.
- 🔗 **Composio** - Use **Composio** to connect 100+ tools. Zero setup. Auth built-in. Made for agents, works for humans.
- 🔗 **Confluent** - Interact with Confluent Kafka and Confluent Cloud REST APIs.
- 🔗 **Contrast Security** - Brings Contrast's vulnerability and SCA data into your coding agent to quickly remediate vulnerabilities.
- 🔗 **Convex** - Introspect and query your apps deployed to Convex.
- 🔗 **Cortex** - Official MCP server for **Cortex**.
- 🔗 **Couchbase** - Interact with the data stored in Couchbase clusters.
- 🔗 **CRIC Wuye AI** - Interact with capabilities of the CRIC Wuye AI platform, an intelligent assistant specifically for the property management industry.
- 🔗 **CrowdStrike Falcon** - Connects AI agents with the CrowdStrike Falcon platform for intelligent security analysis, providing programmatic access to detections, incidents, behaviors, threat intelligence, hosts, vulnerabilities, and identity protection capabilities.
- 🔗 **CTERA Edge Filer** - CTERA Edge Filer delivers intelligent edge caching and multiprotocol file access, enabling fast, secure access to files across core and remote sites.
- 🔗 **CTERA Portal** - CTERA Portal is a multi-tenant, multi-cloud platform that delivers a global namespace and unified management across petabytes of distributed content.
- 🔗 **Cyccode** - Boost security in your dev lifecycle via SAST, SCA, Secrets & IaC scanning with **Cyccode**.
- 🔗 **Dart** - Interact with task, doc, and project data in **Dart**, an AI-native project management tool
- 🔗 **Databricks** - Connect to data, AI tools & agents, and the rest of the Databricks platform using turnkey managed MCP servers. Or, host your own custom MCP servers within the Databricks security and data governance boundary.
- 🔗 **DataHub** - Search your data assets, traverse data lineage, write SQL queries, and more using **DataHub** metadata.
- 🔗 **Daytona** - Fast and secure execution of your AI generated code with **Daytona** sandboxes
- 🔗 **Debugg.AI** - Zero-Config, Fully AI-Managed End-to-End Testing for any code gen platform via **Debugg.AI** remote browsing test agents.
- 🔗 **DeepL** - Translate or rewrite text with **DeepL**'s very own AI models using **the DeepL API**
- 🔗 **Defang** - Deploy your project to the cloud seamlessly with the **Defang** platform without leaving your integrated development environment
- 🔗 **Detailer** - Instantly generate rich, AI-powered documentation for your GitHub repositories. Designed for AI agents to gain deep project context before taking action.
- 🔗 **DevCycle** - Create and monitor feature flags using natural language in your AI coding assistant.
- 🔗 **DevHub** - Manage and utilize website content within the **DevHub** CMS platform
- 🔗 **DevRev** - An MCP server to integrate with DevRev APIs to search through your DevRev Knowledge Graph where objects can be imported from diff. Sources listed **here**.
- 🔗 **DexPaprika (CoinPaprika)** - Access real-time DEX data, liquidity pools, token information, and trading analytics across multiple blockchain networks with **DexPaprika** by CoinPaprika.
- 🔗 **Dolt** - The official MCP server for version-controlled **Dolt** databases.
- 🔗 **Dot (GetDot.ai)** - Fetch, analyze or visualize data from your favorite database or data warehouse (Snowflake, BigQuery, Redshift, Databricks, Clickhouse, ...) with **Dot**, your AI Data Analyst. This remote MCP server is a one-click integration for user that have setup Dot.
- 🔗 **Drata** - Get hands-on with our experimental MCP server—bringing real-time compliance intelligence into your AI workflows.
- 🔗 **Dumpling AI** - Access data, web scraping, and document conversion APIs by **Dumpling AI**

- 🔗 **GitGuardian** - GitGuardian official MCP server - Scan projects using GitGuardian's industry-leading API, which features over 500 secret detectors to prevent credential leaks before they reach public repositories. Resolve security incidents directly with rich contextual data for rapid, automated remediation.
- 🔗 **GitHub** - GitHub's official MCP Server.
- 🔗 **GitKraken** - A CLI for interacting with GitKraken APIs. Includes an MCP server via **gk mcp** that not only wraps GitKraken APIs, but also Jira, GitHub, GitLab, and more.
- 🔗 **Glean** - Enterprise search and chat using Glean's API.
- 🔗 **Globalping** - Access a network of thousands of probes to run network commands like ping, traceroute, mtr, http and DNS resolve.
- 🔗 **gNucleus Text-To-CAD** - Generate CAD parts and assemblies from text using gNucleus AI models.
- 🔗 **Google Cloud Run** - Deploy code to Google Cloud Run
- 🔗 **GoLogin MCP server** - Manage your GoLogin browser profiles and automation directly through AI conversations!
- 🔗 **Google Maps Platform Code Assist** - Ground agents on fresh, official documentation and code samples for optimal geo-related guidance and code..
- 🔗 **gotoHuman** - Human-in-the-loop platform - Allow AI agents and automations to send requests for approval to your **gotoHuman** inbox.
- 🔗 **Grafana** - Search dashboards, investigate incidents and query datasources in your Grafana instance
- 🔗 **Grafbase** - Turn your GraphQL API into an efficient MCP server with schema intelligence in a single command.
- 🔗 **Grain** - Access your Grain meetings notes & transcripts directly in claude and generate reports with native Claude Prompts.
- 🔗 **Graphlit** - Ingest anything from Slack to Gmail to podcast feeds, in addition to web crawling, into a searchable **Graphlit** project.
- 🔗 **Gremlin** - The official **Gremlin** MCP server. Analyze your reliability posture, review recent tests and chaos engineering experiments, and create detailed reports.
- 🔗 **GreptimeDB** - Provides AI assistants with a secure and structured way to explore and analyze data in **GreptimeDB**.
- 🔗 **GROWI** - Official MCP Server to integrate with GROWI APIs.
- 🔗 **Gyazo** - Search, fetch, upload, and interact with Gyazo images, including metadata and OCR data.
- 🔗 **Harper** - An MCP server providing an interface for MCP clients to access data within **Harper**.
- 🔗 **Heroku** - Interact with the Heroku Platform through LLM-driven tools for managing apps, add-ons, dynos, databases, and more.
- 🔗 **HeyOnCall** - Page a human, sending critical or non-critical alerts to the free **HeyOnCall** iOS or Android apps.
- 🔗 **Hiveflow** - Create, manage, and execute agentic AI workflows directly from your assistant.
- 🔗 **Hive Intelligence** - Ultimate cryptocurrency MCP for AI assistants with unified access to crypto, DeFi, and Web3 analytics
- 🔗 **Hologres** - Connect to a **Hologres** instance, get table metadata, query and analyze data.
- 🔗 **Homebrew** Allows **Homebrew** users to run Homebrew commands locally.
- 🔗 **Honeycomb** Allows **Honeycomb** Enterprise customers to query and analyze their data, alerts, dashboards, and more; and cross-reference production behavior with the codebase.
- 🔗 **HubSpot** - Connect, manage, and interact with **HubSpot** CRM data
- 🔗 **Hugging Face** - Connect to the Hugging Face Hub APIs programmatically: semantic search for spaces and papers, exploration of datasets and models, and access to all compatible MCP Gradio tool spaces!
- 🔗 **Hunter** - Interact with the **Hunter API** to get B2B data using natural language.
- 🔗 **Hyperbolic** - Interact with Hyperbolic's GPU cloud, enabling agents and LLMs to view and rent available GPUs, SSH into them, and run GPU-powered workflows for you.
- 🔗 **Hyperbrowser - Hyperbrowser** is the next-generation platform empowering AI agents and enabling effortless, scalable browser automation.
- 🔗 **IBM wxflows** - Tool platform by IBM to build, test and deploy tools for any data source
- 🔗 **Inbox Zero** - AI personal assistant for email **Inbox Zero**
- 🔗 **Inflectra Spira** - Connect to your instance of the SpiraTest, SpiraTeam or SpiraPlan application lifecycle management platform by **Inflectra**
- 🔗 **Inkeep** - RAG Search over your content powered by **Inkeep**
- 🔗 **Integration App** - Interact with any other SaaS applications on behalf of your customers.
- 🔗 **IP2Location.io** - Interact with IP2Location.io API to retrieve the geolocation information for an IP address.
- 🔗 **IPLocate** - Look up IP address geolocation, network information, detect proxies and VPNs, and find abuse contact details using **IPLocate.io**

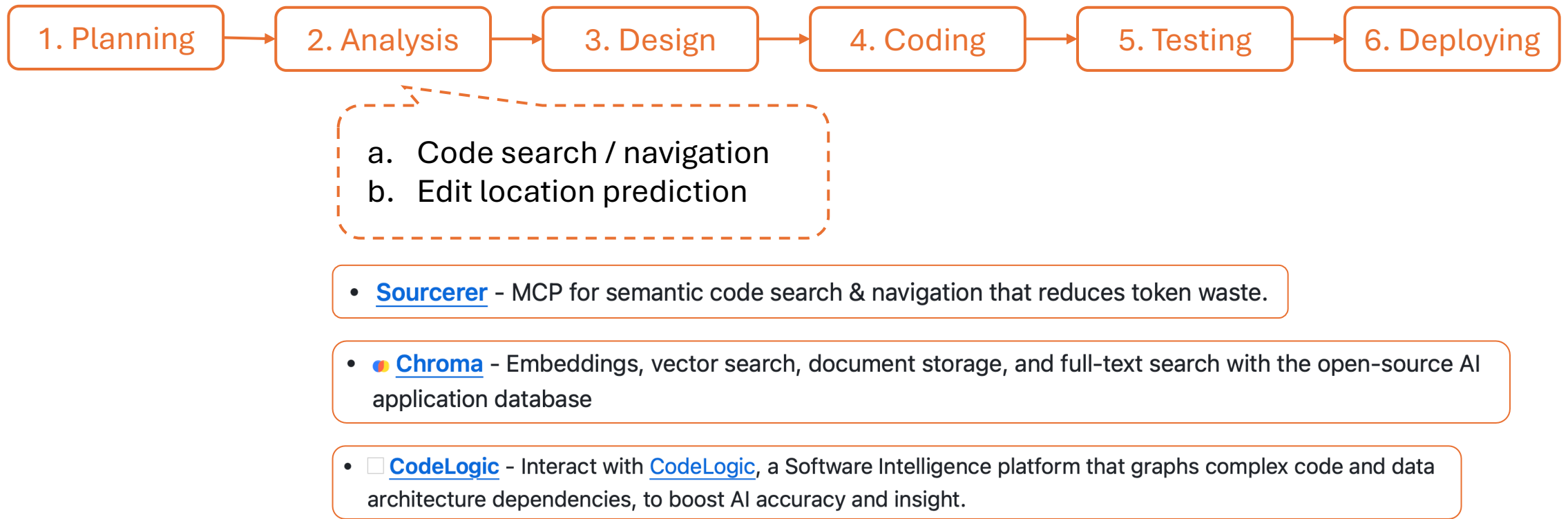
Agentic Framework for Software Development



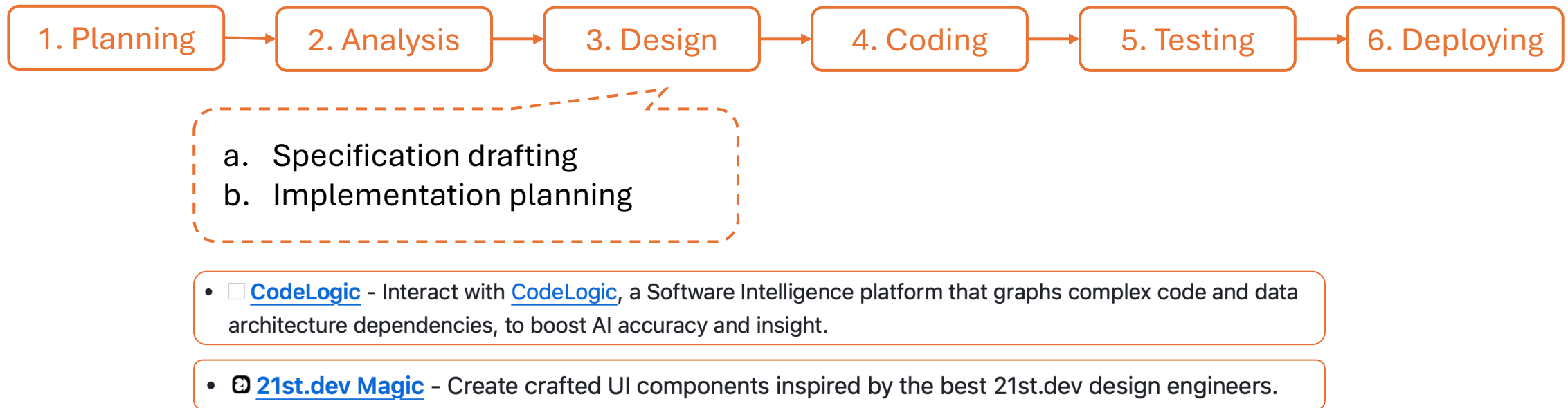
Agentic Framework for Software Development



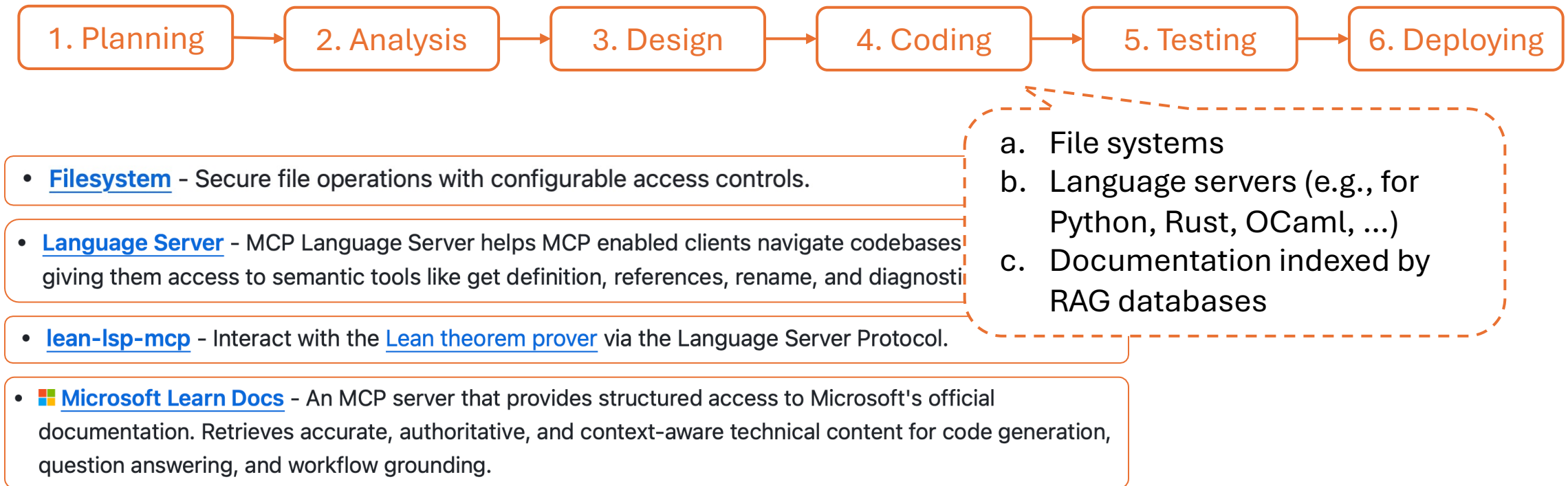
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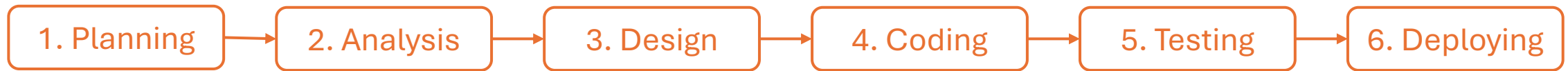
Agentic Framework for Software Development



Agentic Framework for Software Development



Agentic Framework for Software Development



- a. Managing testing environments
- b. Unit testing, regression testing
- c. UI testing, accessibility testing
- d. Vulnerability detection

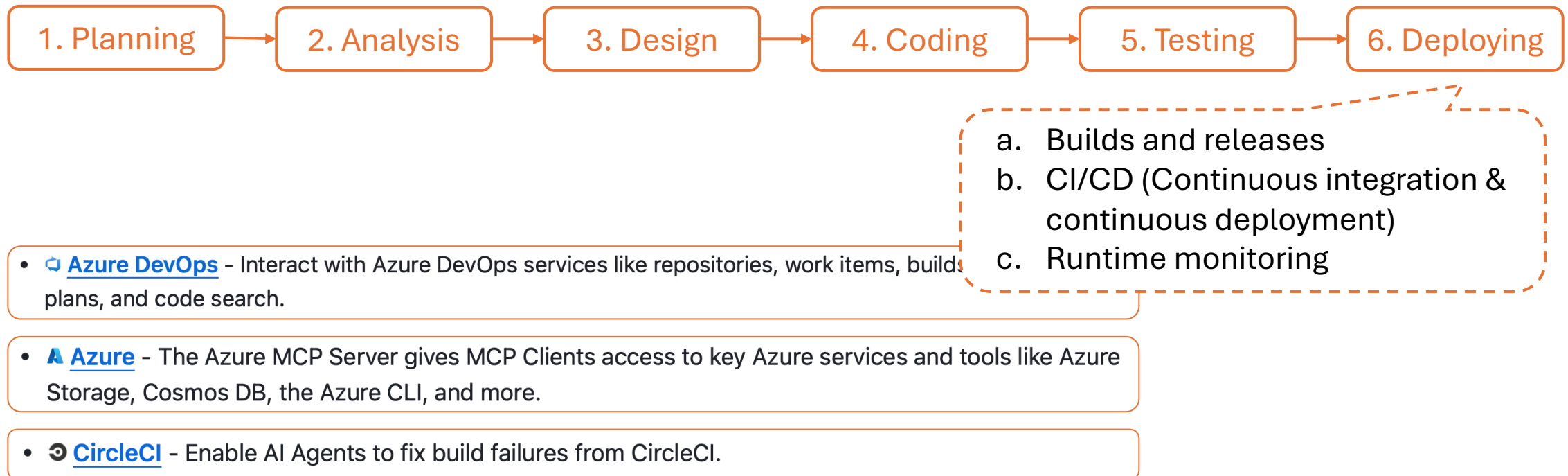
-  [Azure DevOps](#) - Interact with Azure DevOps services like repositories, work items, build plans, and code search.

-  [BrowserStack](#) - Access BrowserStack's [Test Platform](#) to debug, write and fix tests, do accessibility testing and more.

-  [Debugg.AI](#) - Zero-Config, Fully AI-Managed End-to-End Testing for any code gen platform via [Debugg.AI](#) remote browsing test agents.

-  [CrowdStrike Falcon](#) - Connects AI agents with the CrowdStrike Falcon platform for intelligent security analysis, providing programmatic access to detections, incidents, behaviors, threat intelligence, hosts, vulnerabilities, and identity protection capabilities.

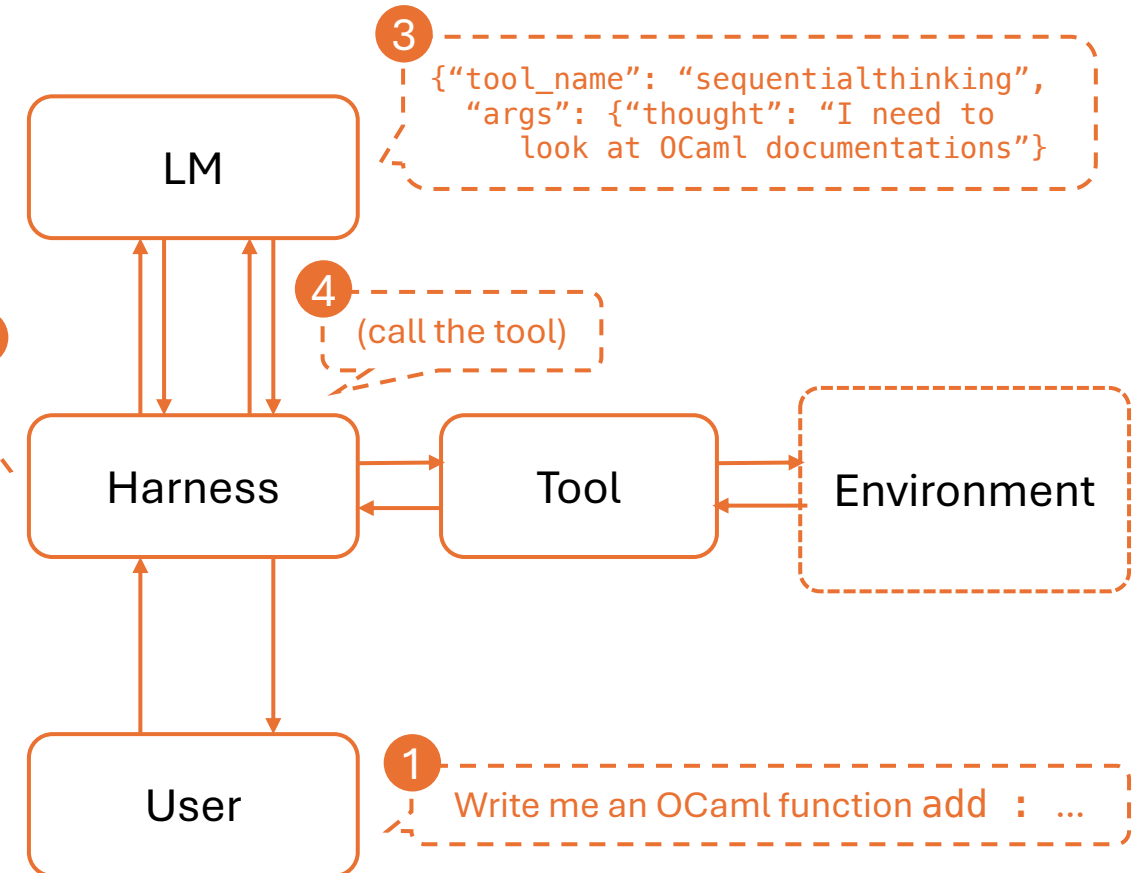
Agentic Framework for Software Development



Sequential Thinking Tool

- **thought** (string)
 - the current step you want to record
- **nextThoughtNeeded** (boolean)
 - whether you inter
- **thoughtNumber** (in
 - index of this step
- **totalThoughts** (int ≥ 1)
 - your *current* plan for how many steps you'll need

The user wants to write a function in OCaml. Here are the available tools: [..., {"name": "sequentialthinking"}]



Tools Related to Software Development

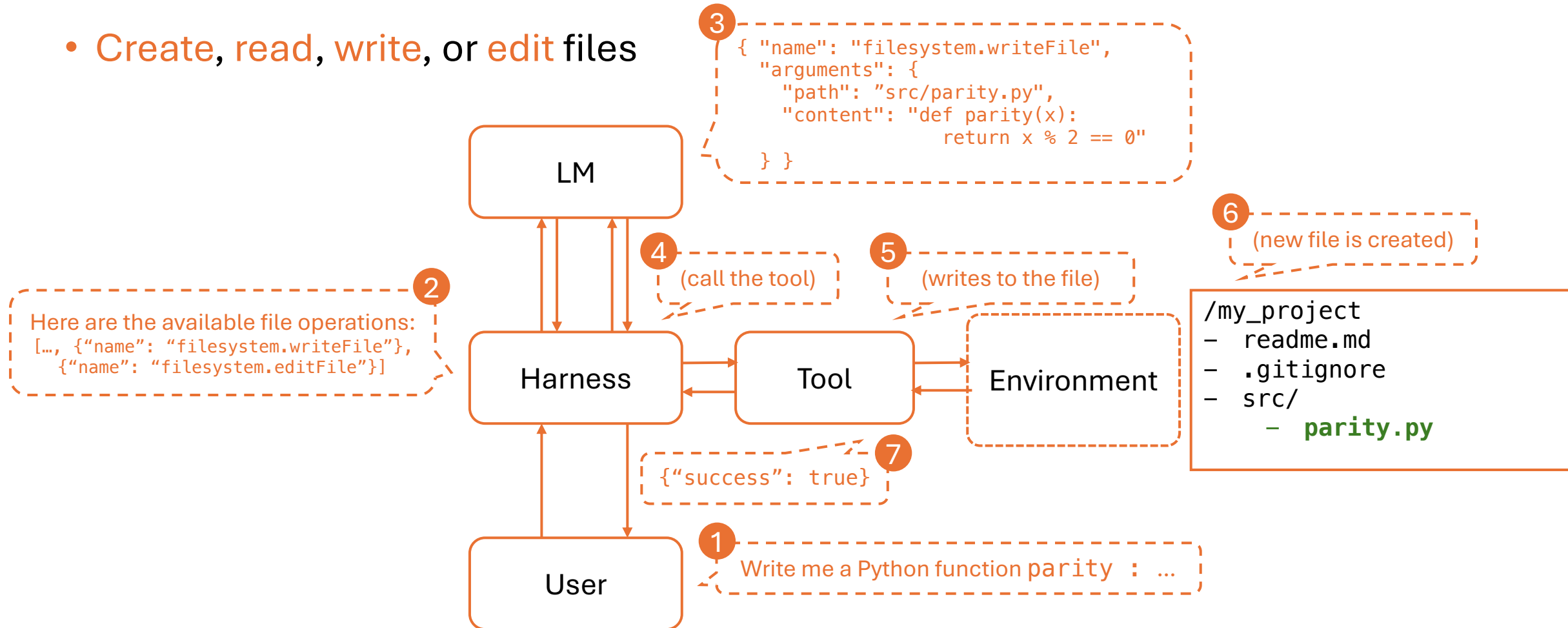
- ~~Sequential thinking~~
 - ~~Planning, deliberate thinking~~
- File systems (FS)
 - Creating, removing, editing programs
 - Drafting test cases and setting up testing environment
- Retrieval augmented generation (RAG) databases
 - Context management
 - Documentations, examples, code snippets

File System (FS) for Code Implementation

- Create, read, write, or edit files
 - Managing project folder structures
 - Put implementation in relevant files, write test cases, etc.
- Search for files
 - Good when we do not know what exact files to work on
 - Often substituted by command line tools such as “grep”

File System (FS) for Code Implementation

- Create, read, write, or edit files



```

{
  name: "read_text_file",
  description:
    "Read the complete contents of a file from the file system as text. " +
    "Handles various text encodings and provides detailed error messages " +
    "if the file cannot be read. Use this tool when you need to examine " +
    "the contents of a single file. Use the 'head' parameter to read only " +
    "the first N lines of a file, or the 'tail' parameter to read only " +
    "the last N lines of a file. Operates on the file as text regardless of extension. " +
    "Only works within allowed directories.",
  inputSchema: zodToJsonSchema(ReadTextFileArgsSchema) as ToolInput,
},

```

```

{

```

```

{
  name: "create_directory",
  description:
    "Create a new directory or ensure a directory exists. Can create multiple " +
    "nested directories in one operation. If the directory already exists, " +
    "this operation will succeed silently. Perfect for setting up directory " +
    "structures for projects or ensuring required paths exist. Only works within allowed directories.",
  inputSchema: zodToJsonSchema(CreateDirectoryArgsSchema) as ToolInput,
},

```

```

    ing file with new content. " +
    es without warning. " +
    rks within allowed directories.",
    ToolInput,
  },
},

```

```

{
  name: "directory_tree",
  description:
    "Get a recursive tree view of files and directories as a JSON structure. " +
    "Each entry includes 'name', 'type' (file/directory), and 'children' for directories. " +
    "Files have no children array, while directories always have a children array (which may be empty). " +
    "The output is formatted with 2-space indentation for readability. Only works within allowed directories.",
  inputSchema: zodToJsonSchema(DirectoryTreeArgsSchema) as ToolInput,
},

```

File System (FS) for Code Implementation

- **Edit file**: how do we specify the “edit” action?
 - Choice 1: Rewrite the entire file
 - Will waste a lot of tokens during iterative refinement
 - **Edit**: new file content **S**
 - Choice 2: Rewrite given lines by line numbers
 - Need to also display “line numbers” when reading the file
 - **Edit**: change lines **X** to **Y** by the string **S**
 - Choice 3: Replace a string with another string
 - Is a bit fragile; the pattern string must be chosen carefully:
 - If it’s too generic, the tool may unintentionally modify multiple occurrences.
 - If there is no exact match, nothing can be edited.
 - **Edit**: replace string **X** with string **Y**

File System (FS) for Code Implementation

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SWE-agent: Agent-Computer Interfaces Enable Automated Software Engineering

John Yang* Carlos E. Jimenez* Alexander Wettig Kilian Lieret
Shunyu Yao Karthik Narasimhan Ofir Press

Princeton Language and Intelligence, Princeton University

- Choice 2: Rewrite given lines by line numbers
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- Edit file: how

- Choice 1: Replace

- Will waste

- Edit: new f

- Choice 2: Rewrite given lines by line numbers

- Need to also display “line numbers” when reading the file

- Edit: change lines X to Y by the string S

- Choice 3: Replace a string with another string

- Is a bit fragile; the pattern string must be chosen carefully:

- If it’s too generic, the tool may unintentionally modify multiple occurrences.

- If there is no exact match, nothing can be edited.

- Edit: replace string X with string Y

Observation (File Viewer)

```
[File: solvers/diophantine.py (3093 lines total)]  
(400 lines above)  
401: elif total_degree == 3:  
402:     diop_type = "cubic_thue"  
403:  
404: elif (total_degree > 3):  
405:     if all(coeff[k] == 1):  
406:         diop_type = 'general_sum_even'  
407:  
408: if diop_type is not None:  
409:     return var, coeff, diop_type  
(2684 lines below)
```

(a) Observation from the file viewer.

Response (Thought + Edit Action)

We need to add a check to ensure that all powers are the same. Let's edit the code to include this check.

```
edit 404:407  
    elif (total_degree > 3 and  
          len(set(k.exp[k.is_Pow])) == 1):  
        if all(coeff[k] == 1):  
            diop_type = 'general_sum_even'  
end_of_edit
```

(b) Action using the edit interface.

Figure 3: The file viewer and edit command are integrated. (a) The file viewer shows the agent the open file’s content with line numbers. (b) The agent invokes the edit function to replace lines 404-407 in the open file. After the edit, the file viewer shows the agent the now updated version of the file.

File System (FS) for Code Implementation

- **Edit file**: how do we specify the “edit” action?
 - Choice 1: Rewrite the entire file
 - Will waste a
 - **Edit**: new file content **S**
 - Choice 2: Rewrite given lines by line numbers
 - Need to also display “line numbers” when reading the file
 - **Edit**: change lines **X** to **Y** by the string **S**
 - **Choice 3: Replace a string with another string**
 - Is a bit fragile; the pattern string must be chosen carefully:
 - If it’s too generic, the tool may unintentionally modify multiple occurrences.
 - If there is no exact match, nothing can be edited.
 - **Edit**: replace string **X** with string **Y**

File System (FS) for Code Implementation

- Edit file: how do we specify the “edit” action?

- Choice

- Will

- Edit:

- Choice

- Need

- Edit:

```
102     const EditOperation = z.object({
103       oldText: z.string().describe('Text to search for – must match exactly'),
104       newText: z.string().describe('Text to replace with')
105     });
106
107     const EditFileArgsSchema = z.object({
108       path: z.string(),
109       edits: z.array(EditOperation),
110       dryRun: z.boolean().default(false).describe('Preview changes using git-style diff format')
111     });
```

- Choice 3: Replace a string with another string

- Is a bit fragile; the pattern string must be chosen carefully:
 - If it's too generic, the tool may unintentionally modify multiple occurrences.
 - If there is no exact match, nothing can be edited.
 - Edit: replace string X with string Y

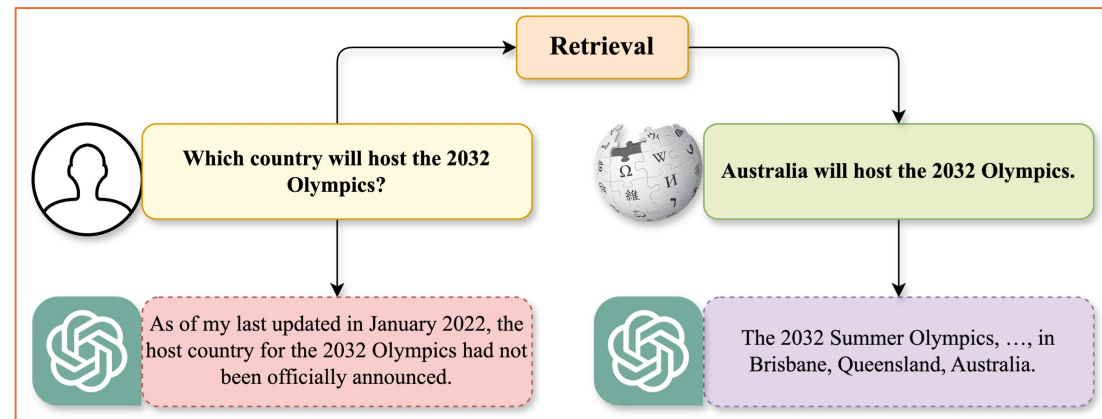
Tools Related to Software Development

- ~~Sequential thinking~~
 - ~~Planning, deliberate thinking~~
- ~~File systems (FS)~~
 - ~~Creating, removing, editing programs~~
 - ~~Drafting test cases and setting up testing environment~~
- Retrieval augmented generation (RAG) databases
 - Context management
 - Documentations, examples, code snippets

Retrieval Augmented Generation (RAG)

- Intuition:

- We do not solely rely on Language Model's internal knowledge
- We assume that there is an **external database** of knowledge
 - Each piece of knowledge is called a “**document**”
- Given a task, we **retrieve** some documents from this database
- Based on the documents, we ask LM to continue **generating** the answers



Retrieval Augmented Generation (RAG)

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

Patrick Lewis^{†‡}, Ethan Perez^{*},

Aleksandra Piktus[†], Fabio Petroni[†], Vladimir Karpukhin[†], Naman Goyal[†], Heinrich Küttler[†],

Mike Lewis[†], Wen-tau Yih[†], Tim Rocktäschel^{†‡}, Sebastian Riedel^{†‡}, Douwe Kiela[†]

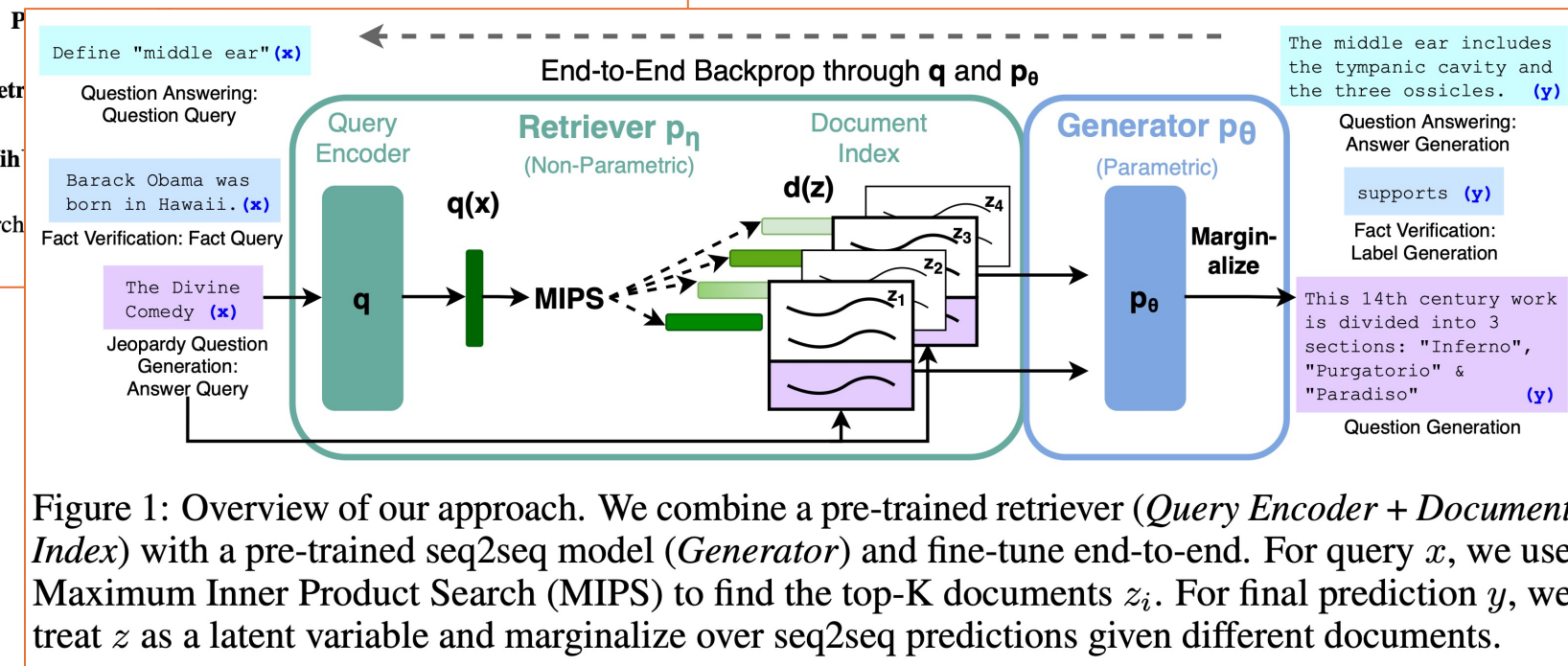
[†]Facebook AI Research; [‡]University College London; ^{*}New York University;
plewis@fb.com

Retrieval-Augmented Generation for Knowledge-Intensive NLP Tasks

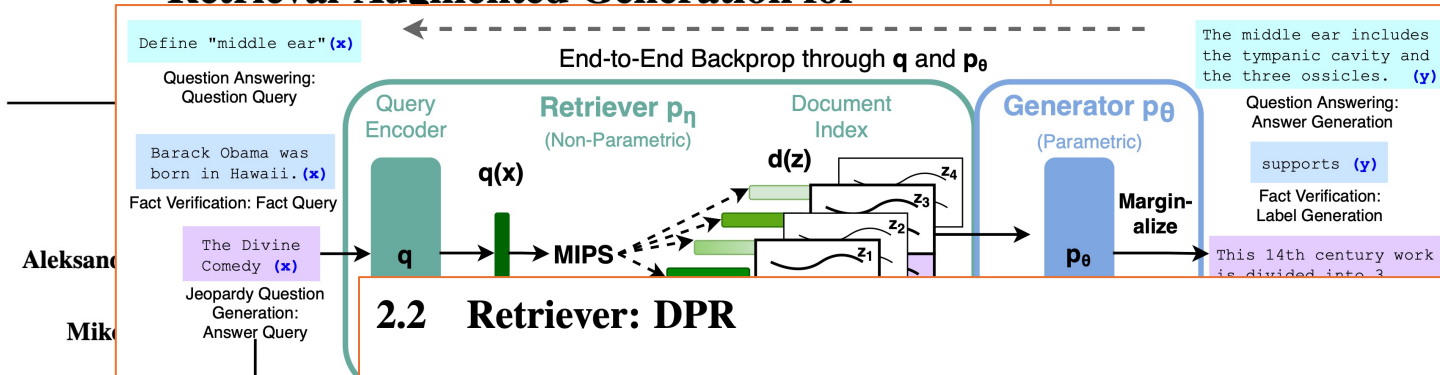
Aleksandra Piktus[†], Fabio Petr

Mike Lewis[†], Wen-tau Yih

[†]Facebook AI Research



Retrieval-Augmented Generation for



2.2 Retriever: DPR

The retrieval component $p_\eta(z|x)$ is based on DPR [26]. DPR follows a bi-encoder architecture:

$$p_\eta(z|x) \propto \exp(\mathbf{d}(z)^\top \mathbf{q}(x)) \quad \mathbf{d}(z) = \text{BERT}_d(z), \quad \mathbf{q}(x) = \text{BERT}_q(x)$$

where $\mathbf{d}(z)$ is a dense representation of a document produced by a $\text{BERT}_{\text{BASE}}$ *document encoder* [8], and $\mathbf{q}(x)$ a query representation produced by a *query encoder*, also based on $\text{BERT}_{\text{BASE}}$. Calculating $\text{top-k}(p_\eta(\cdot|x))$, the list of k documents z with highest prior probability $p_\eta(z|x)$, is a Maximum Inner Product Search (MIPS) problem, which can be approximately solved in sub-linear time [23]. We use a pre-trained bi-encoder from DPR to initialize our retriever and to build the document index. This retriever was trained to retrieve documents which contain answers to TriviaQA [24] questions and Natural Questions [29]. We refer to the document index as the *non-parametric memory*.

The Survey of Retrieval-Augmented Text Generation in Large Language Models

YIZHENG HUANG and JIMMY X. HUANG, York University, Canada

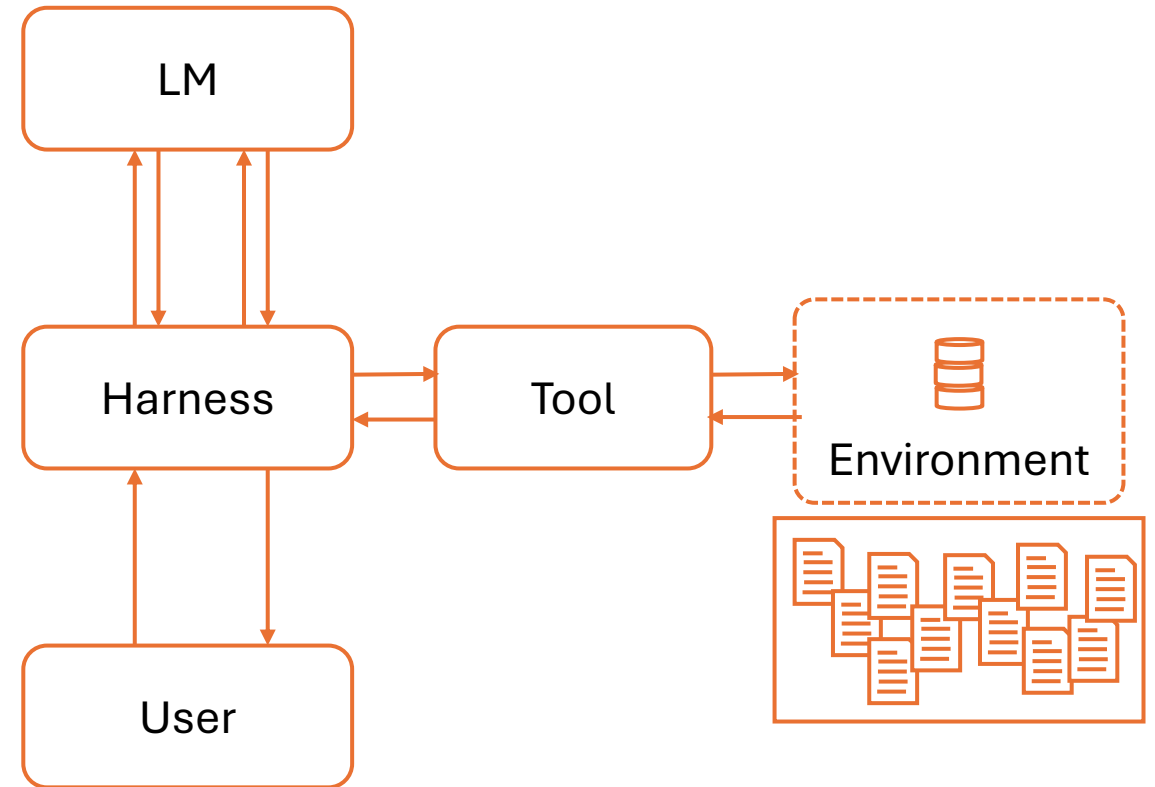
Retrieval-Augmented Generation (RAG) merges retrieval methods with deep learning advancements to address the static limitations of large language models (LLMs) by enabling the dynamic integration of up-to-date external information. This methodology, focusing primarily on the text domain, provides a cost-effective solution to the generation of plausible but possibly incorrect responses by LLMs, thereby enhancing the accuracy and reliability of their outputs through the use of real-world data. As RAG grows in complexity and incorporates multiple concepts that can influence its performance, this paper organizes the RAG paradigm into four categories: pre-retrieval, retrieval, post-retrieval, and generation, offering a detailed perspective from the retrieval viewpoint. It outlines RAG's mechanics and discusses the field's progression through the analysis of significant studies. Additionally, the paper introduces evaluation methods for RAG, addressing the challenges faced and proposing future research directions. By offering an organized framework and categorization, the study aims to consolidate existing research on RAG, clarify its technological underpinnings, and highlight its potential to broaden the adaptability and applications of LLMs.

Retrieval Augmented Generation for Code

- Vector database as environment:
 - Each document z in the database is associated with an embedding $d(z)$
 - Querying: embedding-based similarity search
 - Assume query x , query embedding is $q(x)$
 - Similarity score is cosine similarity between $q(x)$ and $d(z)$:

$$\text{sim}(x, z) = \frac{q(x) \cdot d(z)}{\|q(x)\| \cdot \|d(z)\|}$$

- We can add, update, or remove documents using document metadata



Retrieval Augmented Generation for Code

- Vector database as environment:

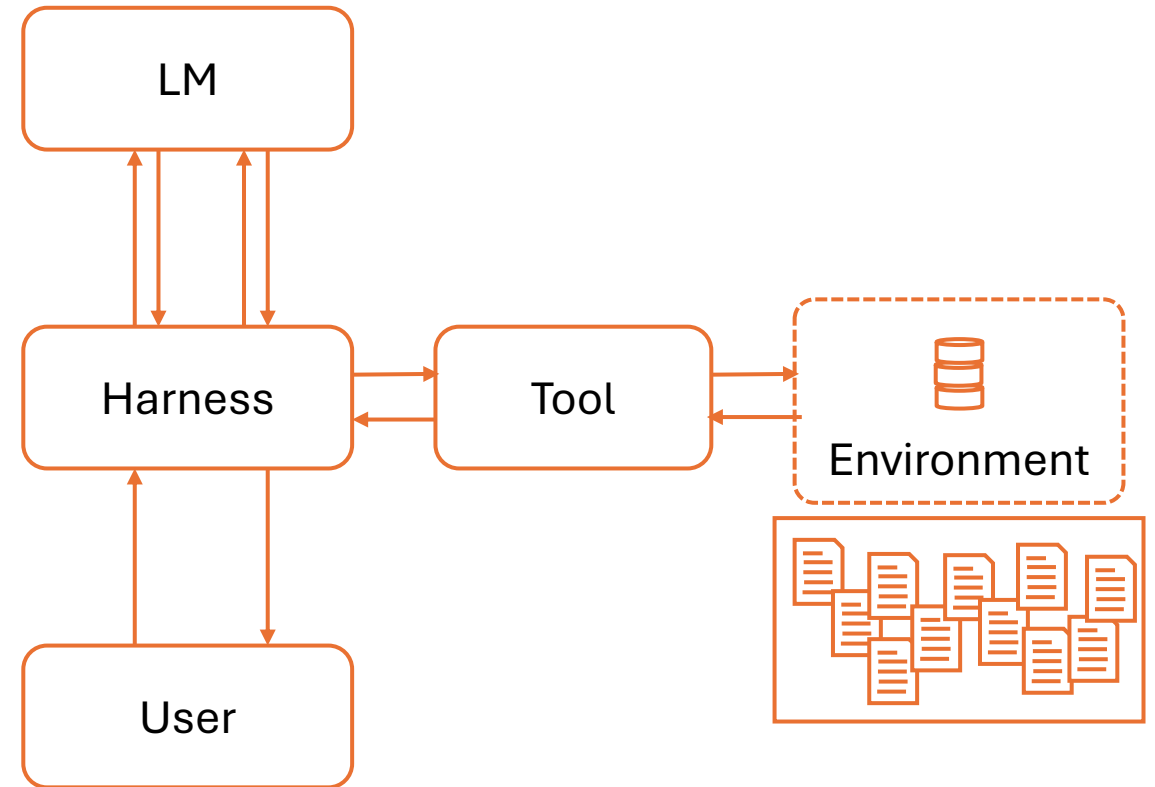
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1

Write me a **Dafny** function Max : ...



Retrieval Augmented Generation for Code

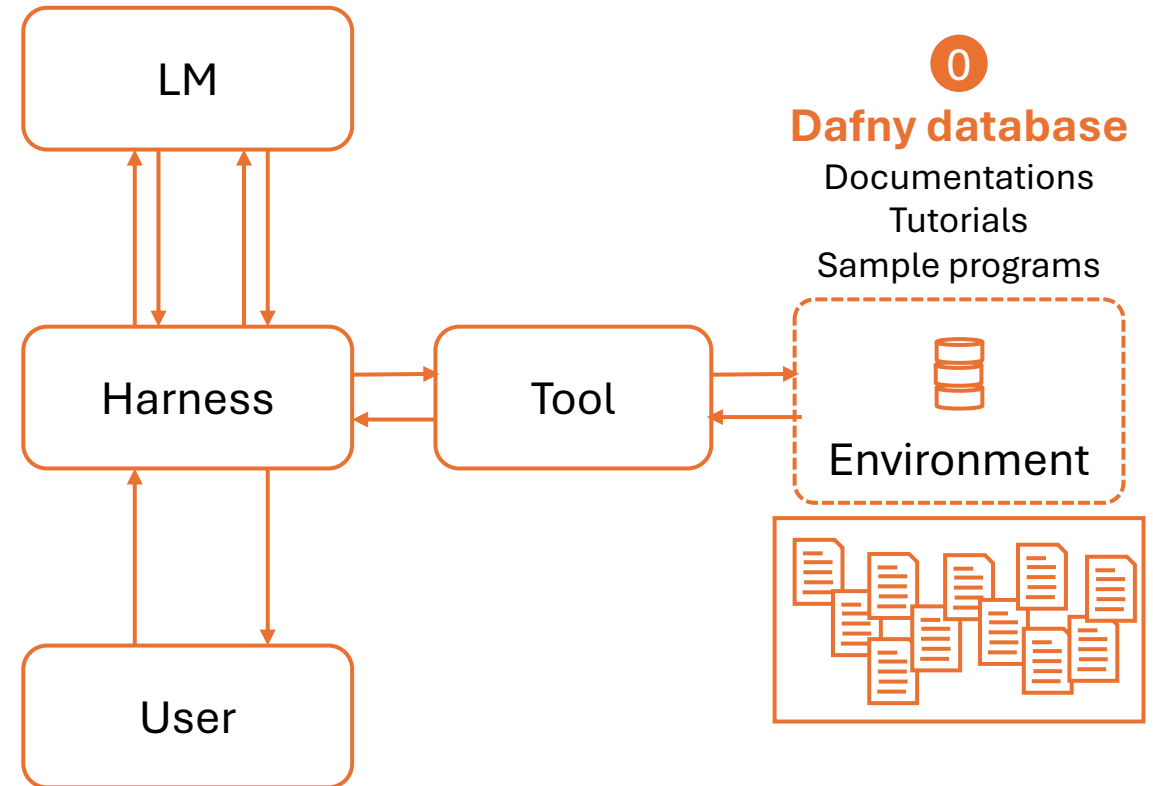
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1 Write me a **Dafny** function Max : ...



Retrieval Augmented Generation for Code

- Vector database as environment:

- Each document z in the database is associated with an embedding $d(z)$
- Querying: embedding-based similarity search
 - Assume query x , query embedding is $q(x)$
 - Similarity score is cosine similarity between

2

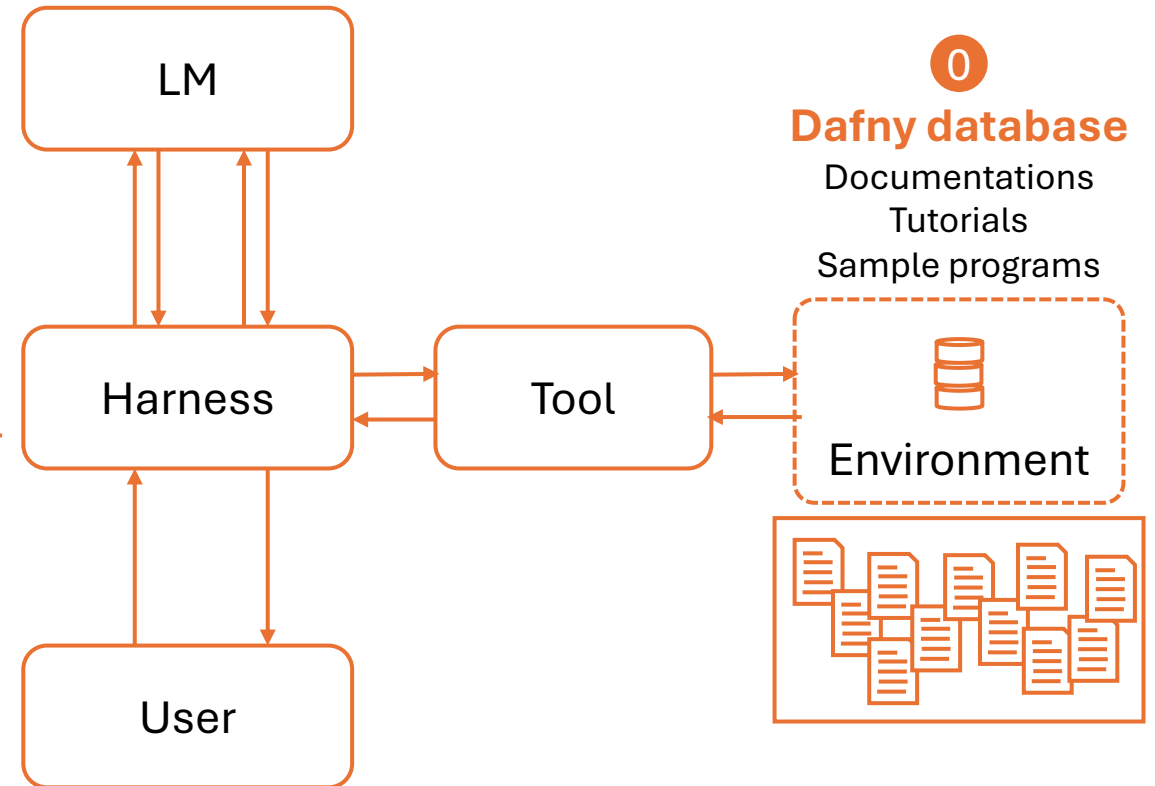
Here are the available tools:
[..., {"name": "chroma.query"}, ...]

$$\text{sim}(x, z) = \frac{q(x) \cdot d(z)}{\|q(x)\| \cdot \|d(z)\|}$$

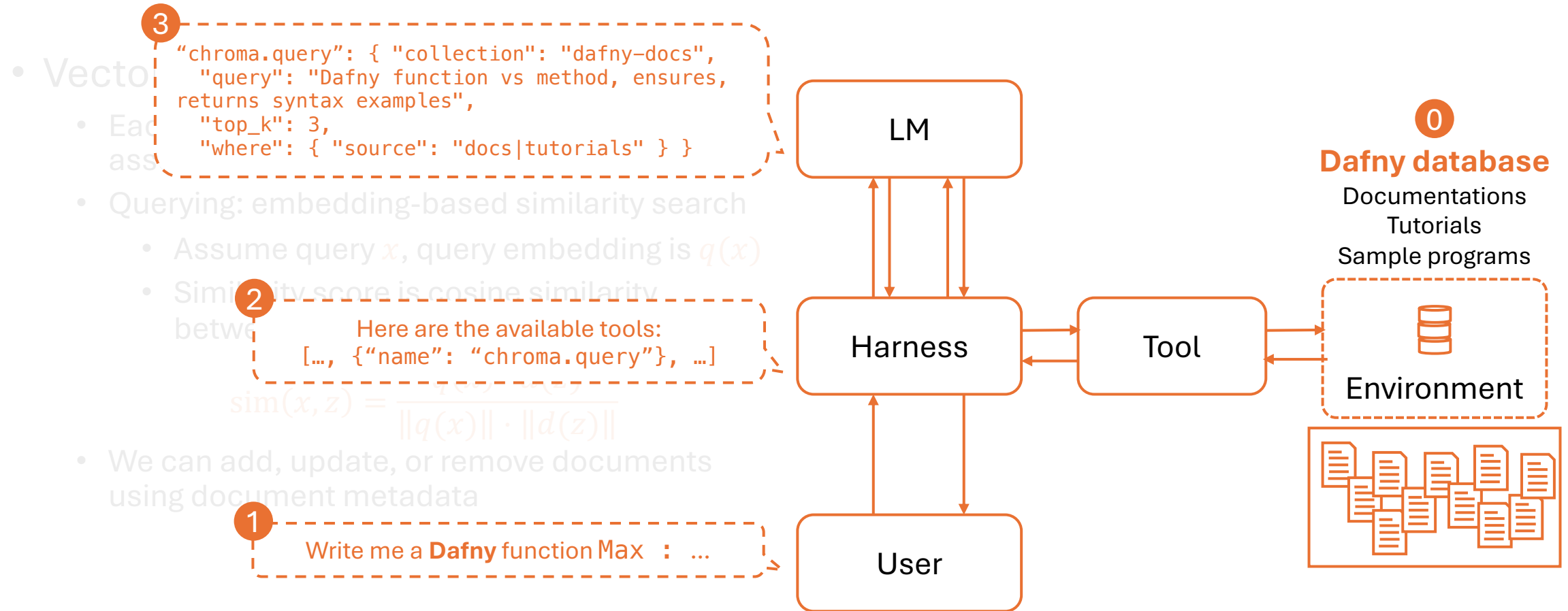
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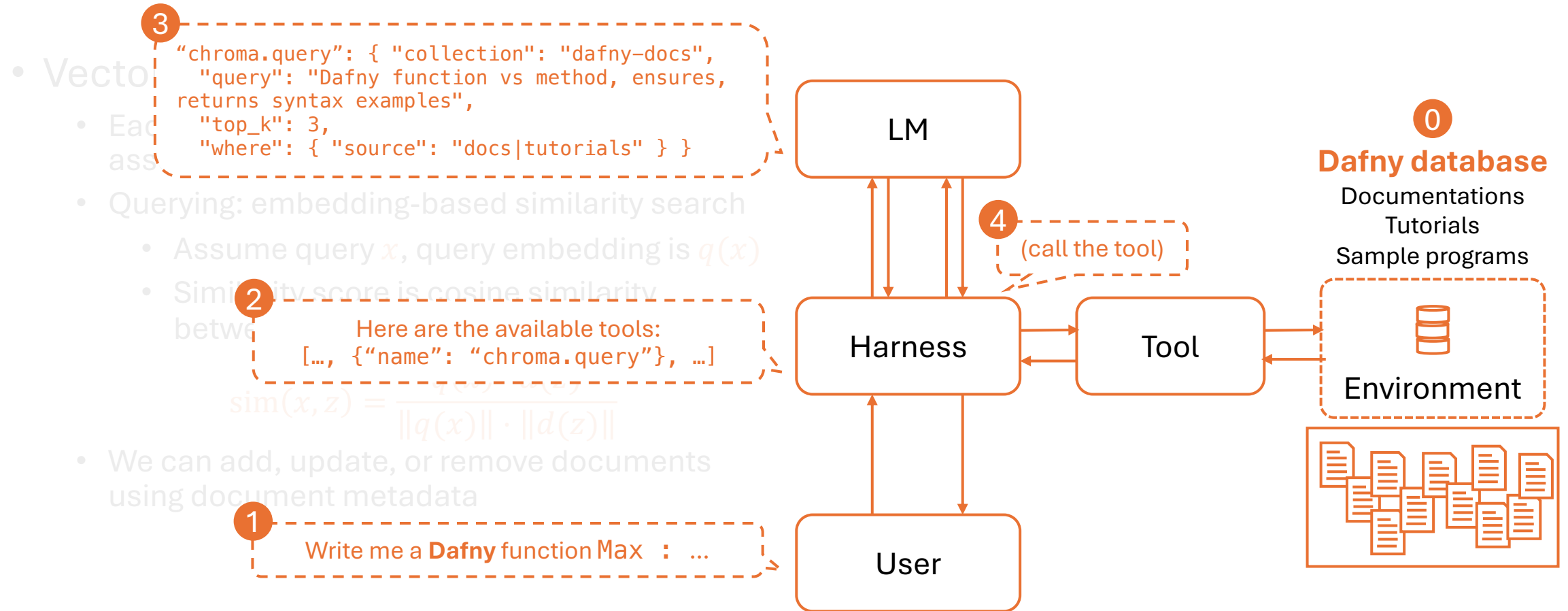
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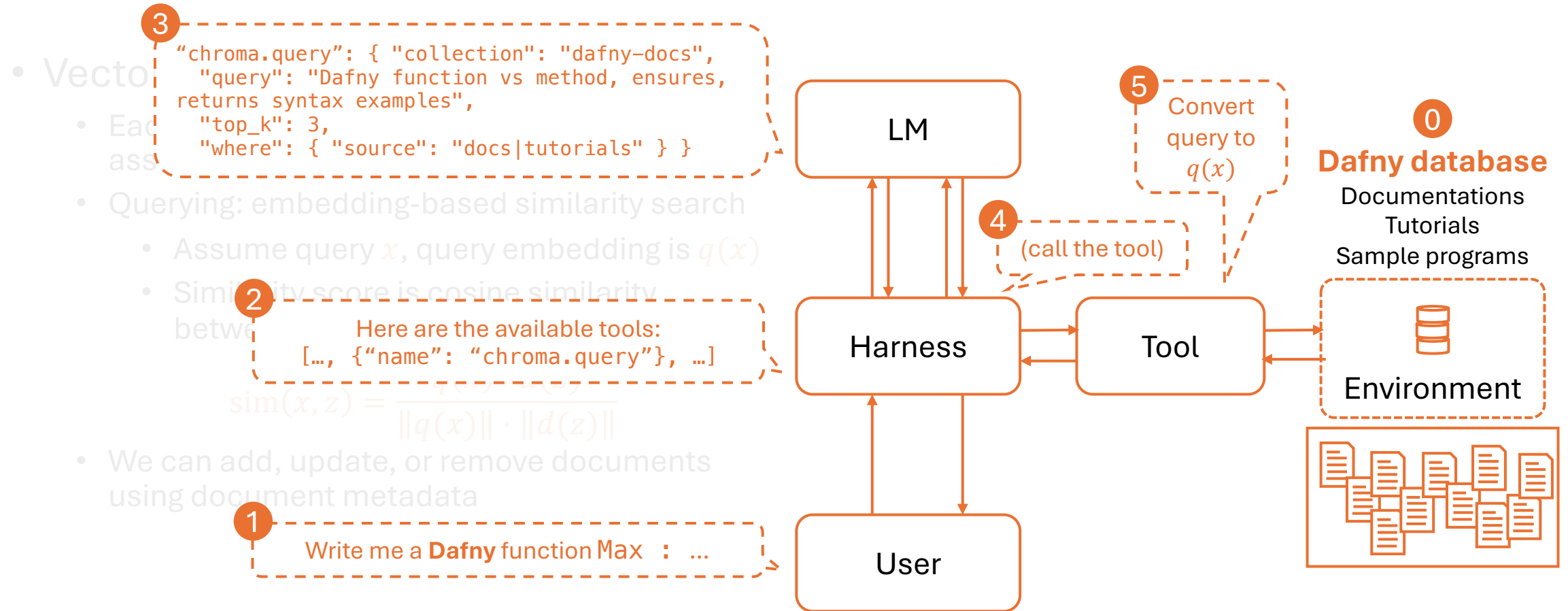
Retrieval Augmented Generation for Code



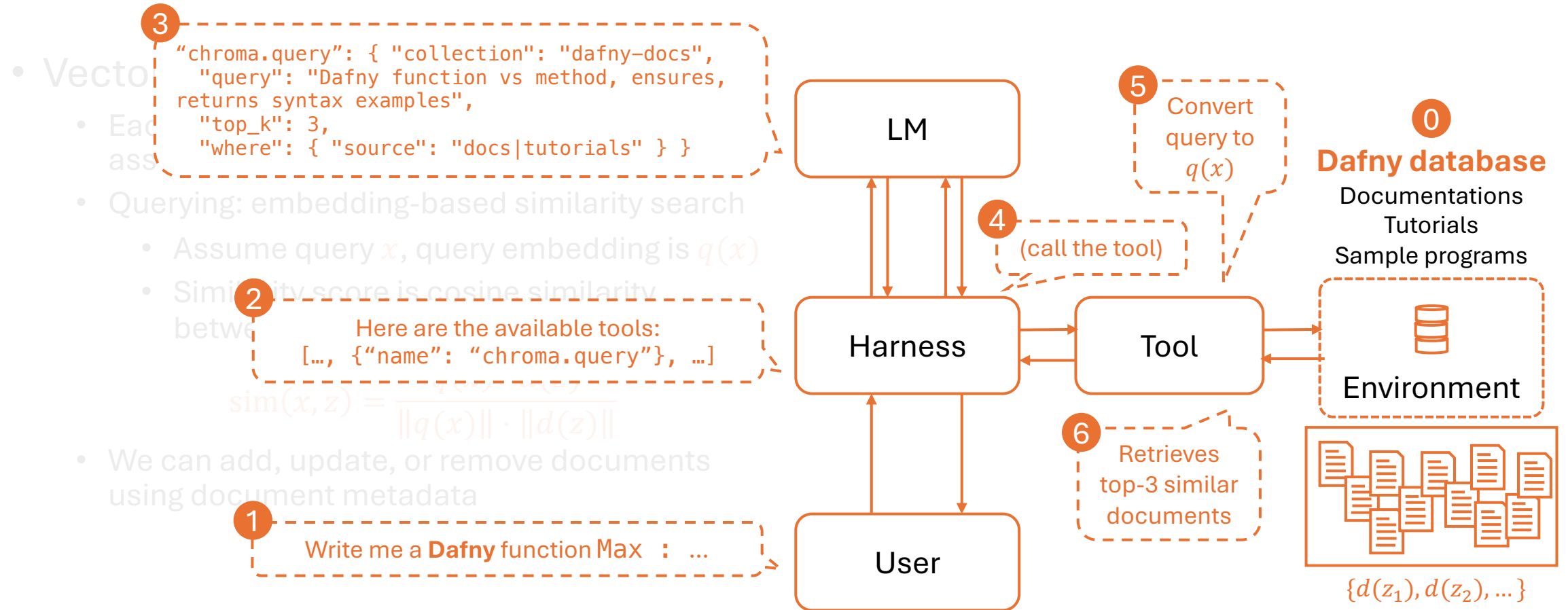
Retrieval Augmented Generation for Code



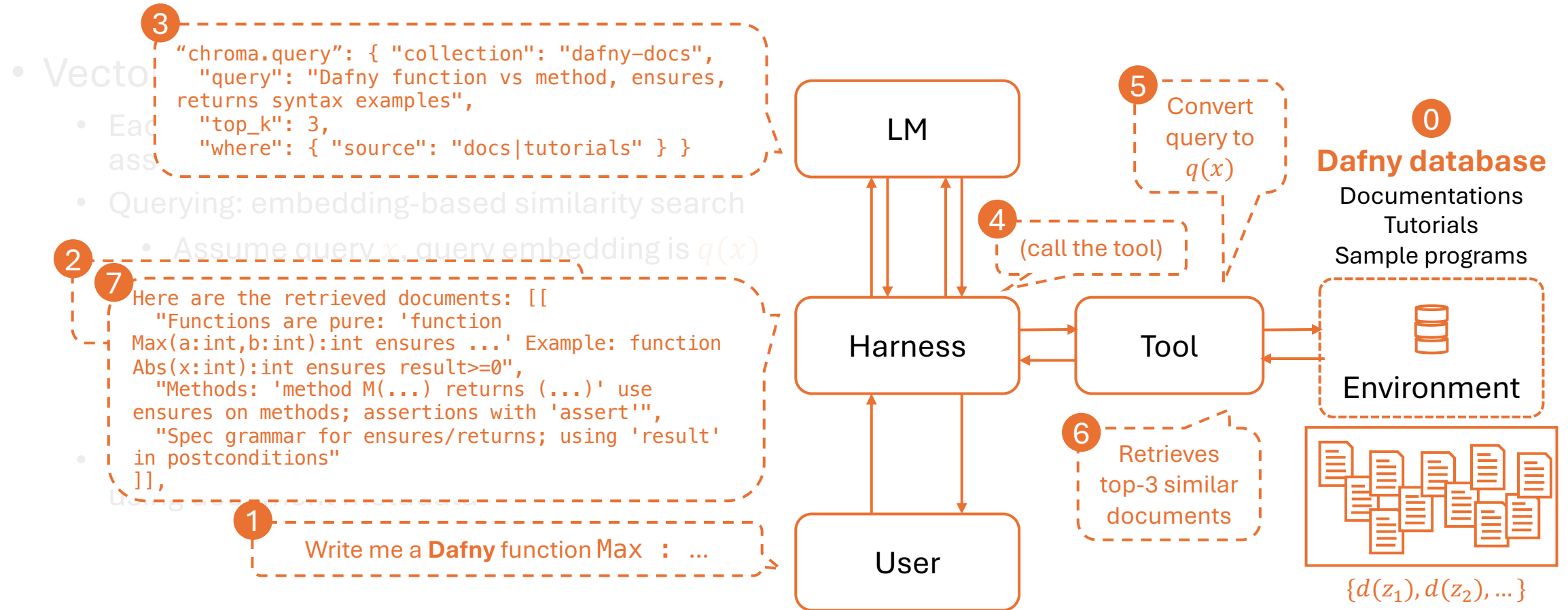
Retrieval Augmented Generation for Code



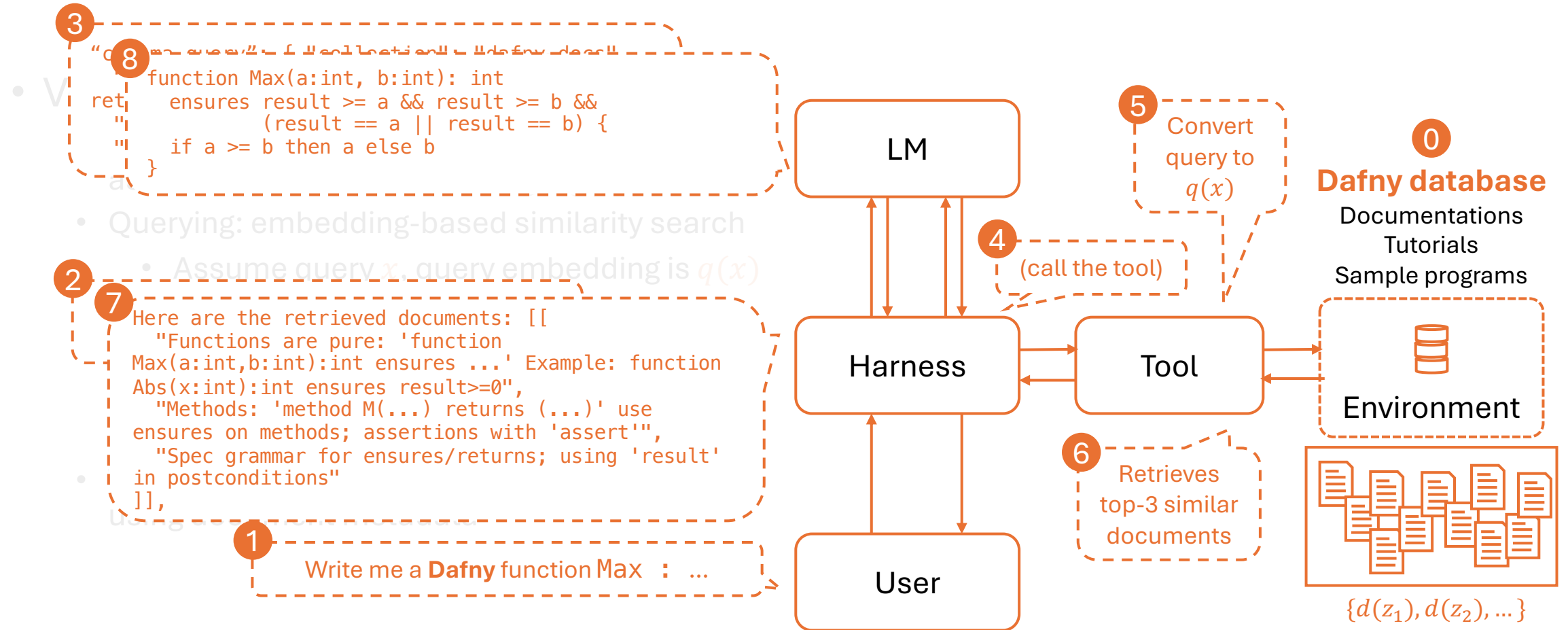
Retrieval Augmented Generation for Code



Retrieval Augmented Generation for Code



Retrieval Augmented Generation for Code

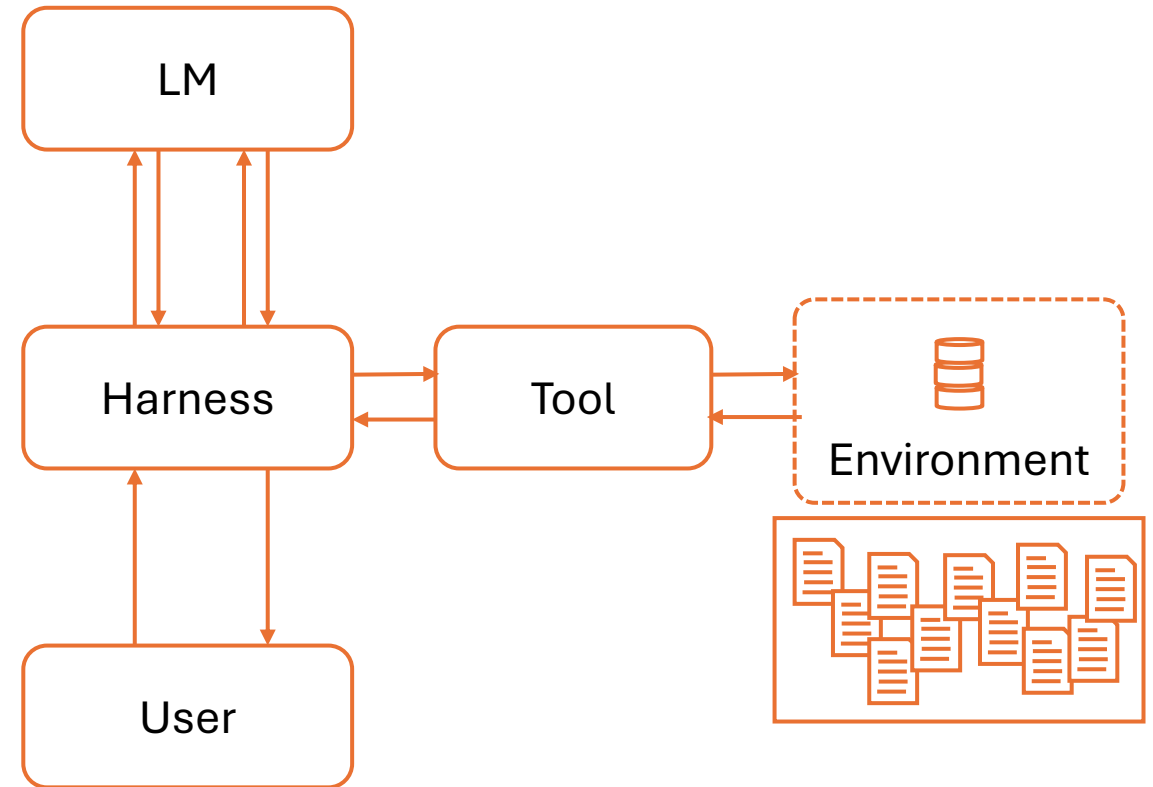


Retrieval Augmented Generation for Code

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CODERAG-BENCH: Can Retrieval Augment Code Generation?

Zora Zhiruo Wang^{♠*} **Akari Asai**^{◇*}

Xinyan Velocity Yu[♡] **Frank F. Xu**[♠] **Yiqing Xie**[♠]

Graham Neubig[♠] **Daniel Fried**[♠]

[♠]Carnegie Mellon University [◇]University of Washington

[♡]University of Southern California

<https://code-rag-bench.github.io/>

CODERAG-BENCH: Can Retrieval Augment Code Generation?

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Graham

[✉] Carnegie Mellon U

Unive

<https://carnegie Mellon.edu/>

8 Coding Tasks for Code RAG

Basic Programming x 3

```
def has_close_elements(
    numbers: List[float], threshold: float)
    -> bool:
    """ Check if in given list of numbers,
        are any two numbers closer to each
        other than given threshold.
    """
```

Open-Domain x 2

```
df.columns = np.concatenate(
    [df.iloc[0, :2], df.columns[2:]])
df =
df.iloc[1:].reset_index(drop=True)
return df
```

```
import
numpy as
np
import
pandas
```

Repository-Level x 2

```
def main():
    utils.copy_img(data, img_dir)
    utils.downscale(img_dir, config)
    # call COLMAP executable
    >> _____
```

```
run_colmap(
    image_path, ...)
def downscale(
    img_dir, config):
    ...
    return img
```

Code Retrieval x 1

```
Q: Print a log message to
standard error.
def print_log(text, *colors):
    sys.stderr.write(
        sprint(text, *colors))
```

5 Document Sources for Retrieval

Programming Solutions

```
def truncate_number(number) -> float:
    """Return the decimal part. """
    return number % 1.0
```

Library Documentation

```
numpy.dot(a, b, out=None)
Dot product of two arrays.
• If both a and b are 1-D arrays ...
```

Online Tutorials

Example #1: Using shutil.move() to move file from source to destination

```
... import shutil
dst = shutil.move(src, dst_path) ...
```

oneuser/codemodels

StackOverflow Posts

Pandas map multiple columns based on specific conditions

```
# fill NaN values with old values
out["nid"] = out["nid"].fillna(out["id"])
```

GitHub Repositories

Reproducible retrieval and end-to-end evaluation

Models

VOYAGE AI
OpenAI
elasticsearch

starcoder
deepseek coder

Datastore

Retriever

LM

Evaluation

✓ Evaluation against canonical docs

✓ Execution-based end-to-end evaluation

CodeRAG-BENCH:

8 Coding Tasks for Code RAG

Basic Programming x 3

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```

Reproducible retrieval and end-to-end evaluation



Type	Dataset	# Examples	# Corpus	Ground-Truth Docs	Evaluation
Basic programming	HumanEval	164	164	program solutions	execution
	MBPP	500	500	program solutions	execution
	LiveCodeBench	400	-	-	execution
Open-domain	DS-1000	1000	34,003	docs	execution
	ODEX	945	34,003	docs, stackoverflow	execution
Repository-level	RepoEval (function)	373	237	github repository	execution
	SWE-bench-Lite	300	40,868	github repository	execution
Code retrieval	CodeSearchNet-Py	22,177	22177	CSN functions	ndcg@10

Method	Basic Programming			Open-Domain			Repo-Level	
	HumanEval	MBPP	LCB	DS-1000	ODEX	ODEX-hard	RepoEval	SWE-bench
w/ StarCoder2-7B								
None	31.7	2.4	1.5	29.2	14.6	10.3	26.5	0.0
BM25	43.9	51.8	1.0	36.7	14.1	13.8	36.7	0.0
GIST-large	38.7	50.4	0.5	35.9	17.3	13.8	40.8	0.3
Voyage, code	39.0	52.6	0.3	36.0	15.3	10.3	45.8	0.3
OpenAI, small	39.0	52.6	1.5	35.5	15.9	17.2	51.2	0.0
OpenAI, rerank	34.8	53.4	0.5	33.4	14.1	17.2	53.9	0.3
Gold	94.5	34.8	-	30.0	17.5	17.2	42.0	0.7
w/ DeepseekCoder-7B-instruct								
None	70.1	60.8	30.5	41.4	39.2	17.2	28.2	0.7
BM25	68.9	60.0	31.8	36.6	37.8	20.7	37.3	0.0
GIST-large	66.3	56.6	33.8	35.9	34.9	20.7	44.5	0.3
Voyage, code	66.5	56.4	31.8	35.9	39.4	17.2	46.6	0.3
OpenAI, small	68.9	58.6	32.0	35.5	37.1	20.7	55.2	0.3
OpenAI, rerank	53.0	60.6	31.5	36.5	37.1	24.1	55.5	0.3
Gold	87.8	63.6	-	43.2	41.7	24.1	48.1	0.0
w/ GPT-3.5-turbo				GPT-4o				
None	72.6	70.8	35.3	43.7	41.7	17.2	23.9	2.3
BM25	73.2	72.4	35.5	36.9	41.0	24.1	30.8	6.7
GIST-large	73.2	68.2	34.8	36.7	36.2	13.8	38.3	19.3
Voyage, code	75.0	66.8	34.5	37.4	41.0	20.7	43.2	15.7
OpenAI, small	73.8	68.4	35.8	36.9	40.3	17.2	48.0	21.0
OpenAI, rerank	64.0	72.6	33.5	37.4	40.5	17.2	49.6	21.7
Gold	91.5	72.6	-	42.9	40.3	24.1	39.1	30.7

Table 6: Performance of retrieval-augmented code generation, with top retrieval and generation models. We bold-type the best RACG results. We test gpt-4o on SWE-bench to show non-trivial results than gpt-3.5-turbo. Note that we exclude code canonical answer from the retrieval corpora for basic programming tasks.

CODERAG-BENCH:

8 Coding Tasks for Code RAG

Basic Programming x 3

```
def has_close_elements(
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    -> bool:
    """ Check if in
    are any two num
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    """
```

Open-Domain x 2

```
df.columns = np.concatenate(
    [df.columns, df.columns, ...])
```

Repository-Level x 2

```
import numpy as
def main():
    util.copy_image(data_image, dic)
```

Code Retrieval x 1

```
Q: Print a log message to
standard error.
```

Type

Basic programming

Open-domain

Repository

Code retrieval

StackOverflow posts

Method

None

BM25

GIST-large

Voyage, code

OpenAI, small

OpenAI, rerank

Gold

None

BM25

GIST-large

Voyage, code

OpenAI, small

OpenAI, rerank

Gold

Table 6: Performance of different methods. Bold-type the best RAG method that we exclude code completion.

● Tutorials ● Library Documentation
● StackOverflow posts ● Github files

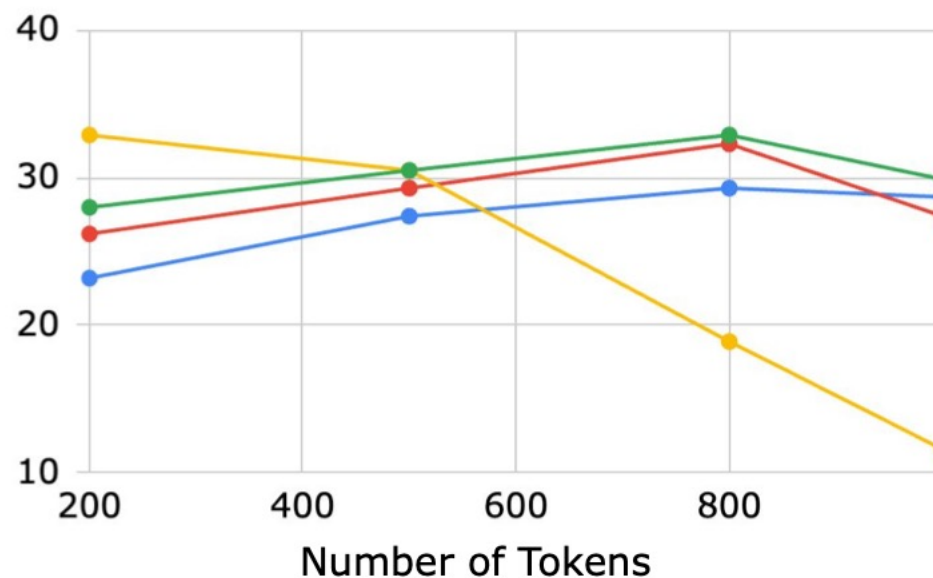


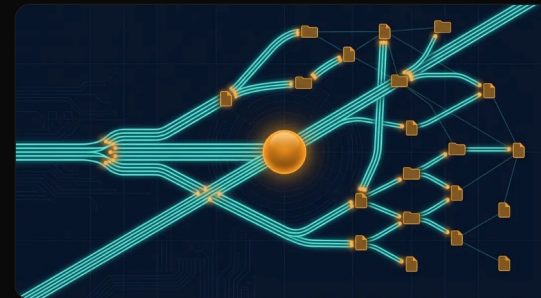
Figure 2: Performance with different chunking sizes.

Note: chunking size is the maximum number of tokens per document; A larger document (e.g., StackOverflow post) will be chunked into multiple smaller documents of chunking size

Agentic Search: How Coding Agents Find the Right Code

Agentic search uses multi-step reasoning and parallel tool calls to find code. How it works, why it beats semantic search and grep, and why it matters for AI coding agents.

February 23, 2026 · 4 min read



Agentic Coding: The Right Choice

Agentic search uses machine learning to understand code context, works, why it beats semantic search, and when to use agents.

February 23, 2026 · 4 min read

Agentic Search vs Semantic Search vs Lexical Search

SEARCH PARADIGM COMPARISON

DIMENSION	SEMANTIC SEARCH	LEXICAL SEARCH	AGENTIC SEARCH
How it works	Embed query + docs, return nearest neighbors	Pattern matching (grep, ripgrep, regex)	Multi-turn reasoning with parallel tool calls
Strengths	Fuzzy matching, conceptual similarity	Exact matches, fast, zero indexing cost	Follows causal chains, multi-file reasoning
Weaknesses	No causal logic, stale embeddings	Misses semantic intent, returns noise	Higher compute cost, needs training
Best for	Document/knowledge-base retrieval	Symbol lookup, known identifiers	Multi-file code understanding
Typical latency	< 100ms	< 50ms	2-8 seconds
Indexing required	Yes (embeddings)	No	No
Handles code structure	No — treats code as flat text	Partially — matches symbols	Yes — follows imports, call chains, data flow

Agentic Coding Right C

Agentic search uses m
works, why it beats ser
agents.

February 23, 2026 · 4

Agentic Search vs Semantic Search vs Lexical Search

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DIMENSION	SEMANTIC SEARCH	LEXICAL SEARCH	AGENTIC SEARCH
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Strengths	Fuzzy matching, conceptual similarity	Exact matches, fast, zero indexing cost	Follows causal chains, multi-file reasoning
Weaknesses	No causal logic, stale embeddings	Misses semantic intent, returns noise	Higher compute cost, needs training
Best for	Document/knowledge-base retrieval	Symbol lookup, known identifiers	Multi-file code understanding
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Knowledge Retrieval

- **Parametric knowledge**
 - Knowledge embedded inside model weights
 - Obtained by
 - Model training (pre-/post-training)
- **Inference-time / Index-time knowledge**
 - Knowledge only accessible
 - Obtained by
 - RAG Vector Databases
 - Terminal search methods (grep, find, etc.)
 - Code actions such as go-to-definition (language-servers, API)
 - Web search

Terminal as Tool

- Usage

- Running **compilers**: `gcc foo.c` → syntax/type errors
- Running **programs**: `python foo.c` → runtime errors
- Running **tests**: `pytest` → test cases pass/fail
- Managing **packages**: `pip` (Python), `cargo` (Rust), `npm` (JavaScript)
- Processing **files**: `cat` (read), `grep/find` (search), `sed` (edit), `echo` (file write)
- Managing **folders** and **directories**: `cd` (go to dir), `ls` (list items in dir)

- Interaction

- Input: **command** (string)
- Output: **stdout** / **stderr** (string), **exit code** (integer)

Now I need to update the test files to use the correct types. Let me update the test that creates generic types:

test_type_inference.rs +1 -1 •

× ✓ ◇

Now let me try to compile again:

Auto-Ran command: cd, cargo check

↗ ◇

```
$ cd /Users/liby99/Local/Projects/scallop-lang/scallop-v3 && cargo check --package scallop-fir
```

```
For more information about an error, try `rustc --explain E0277`.
```

```
warning: `scallop-fir` (lib) generated 1 warning
```

```
error: could not compile `scallop-fir` (lib) due to 4 previous errors; 1 warning emitted
```

I see the issue. The problem is that I'm trying to match on `Type::Named` and `Type::Generic` but the AST `Type` is actually a struct, not an enum. Let me check the actual structure of the AST `Type`:

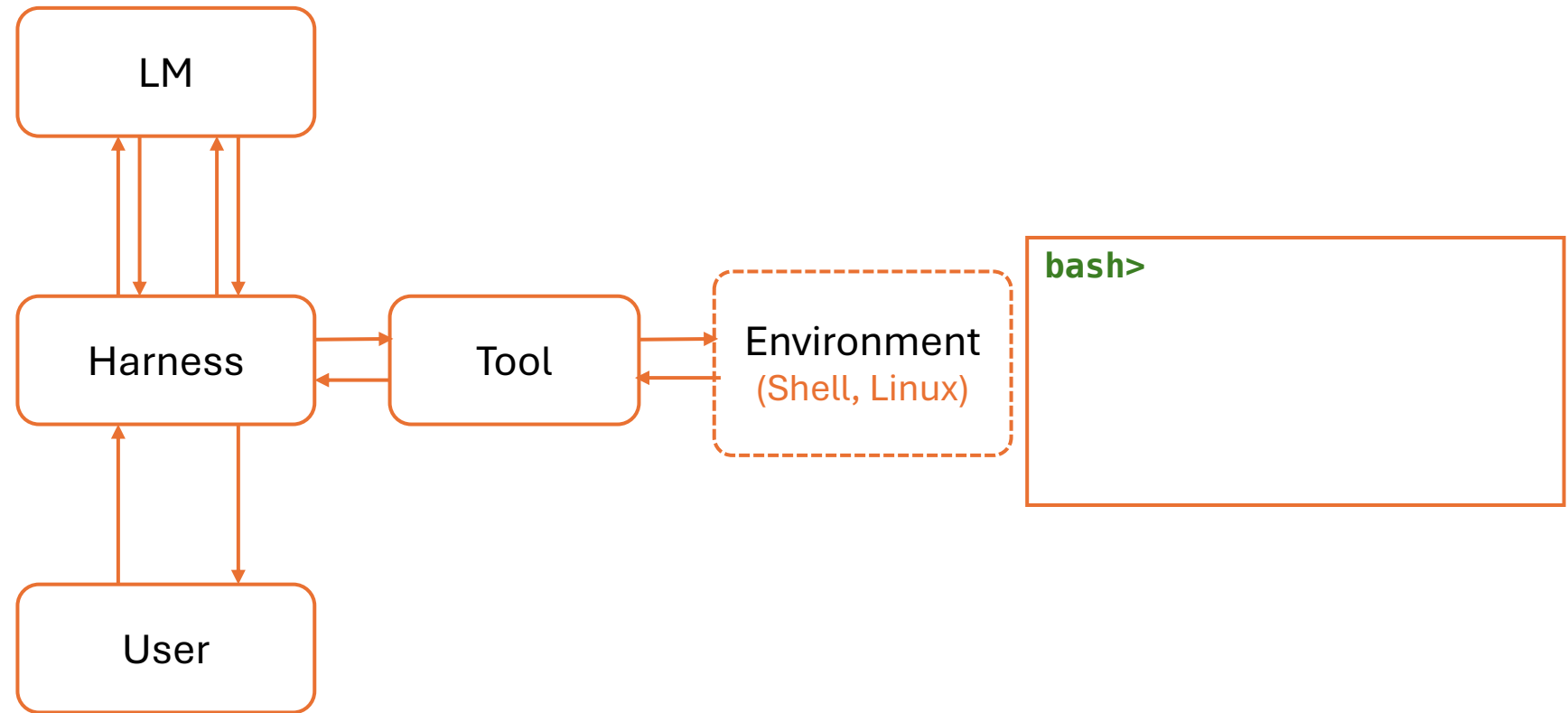
Read types.rs

I see! The AST `Type` is actually a struct that represents both named types and generic types. If `type_args` is empty, it's a named type. If `type_args` is not empty, it's a generic type. Let me fix the type checking method:

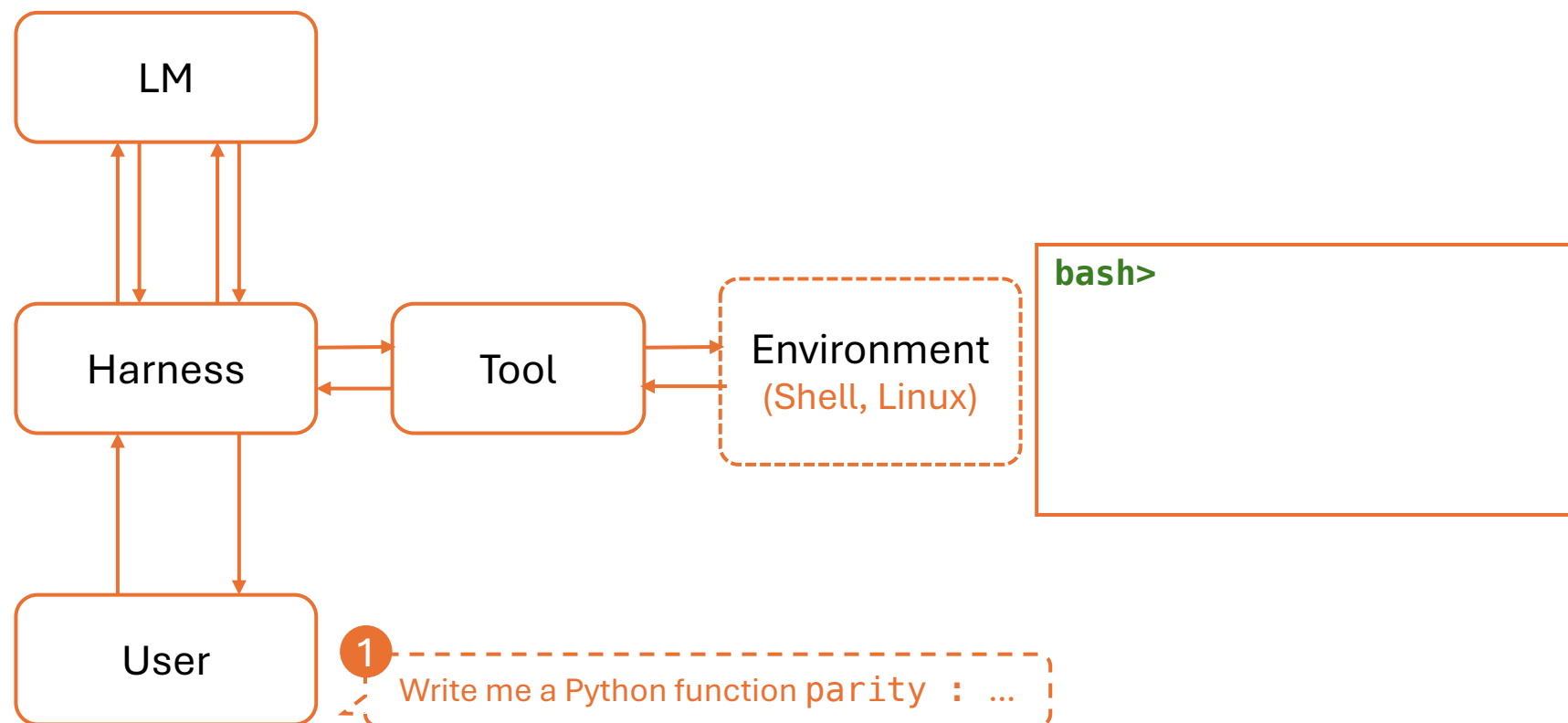
type_inference.rs +10 -18 •

× ✓ ◇

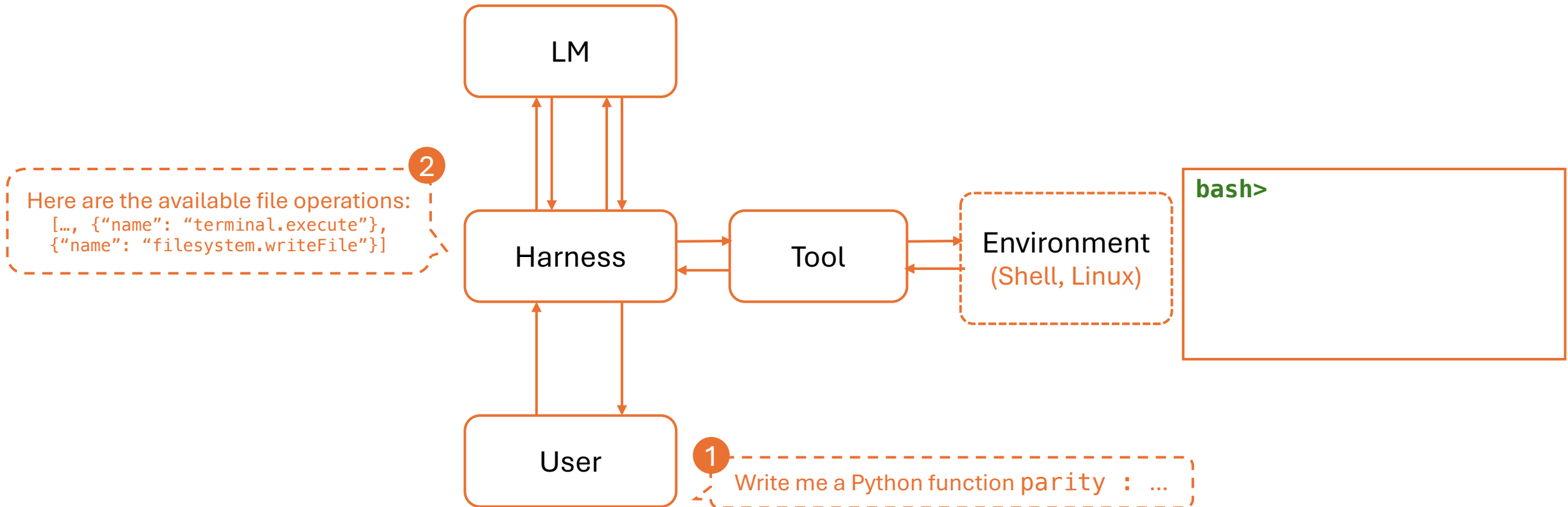
Terminal as Tool



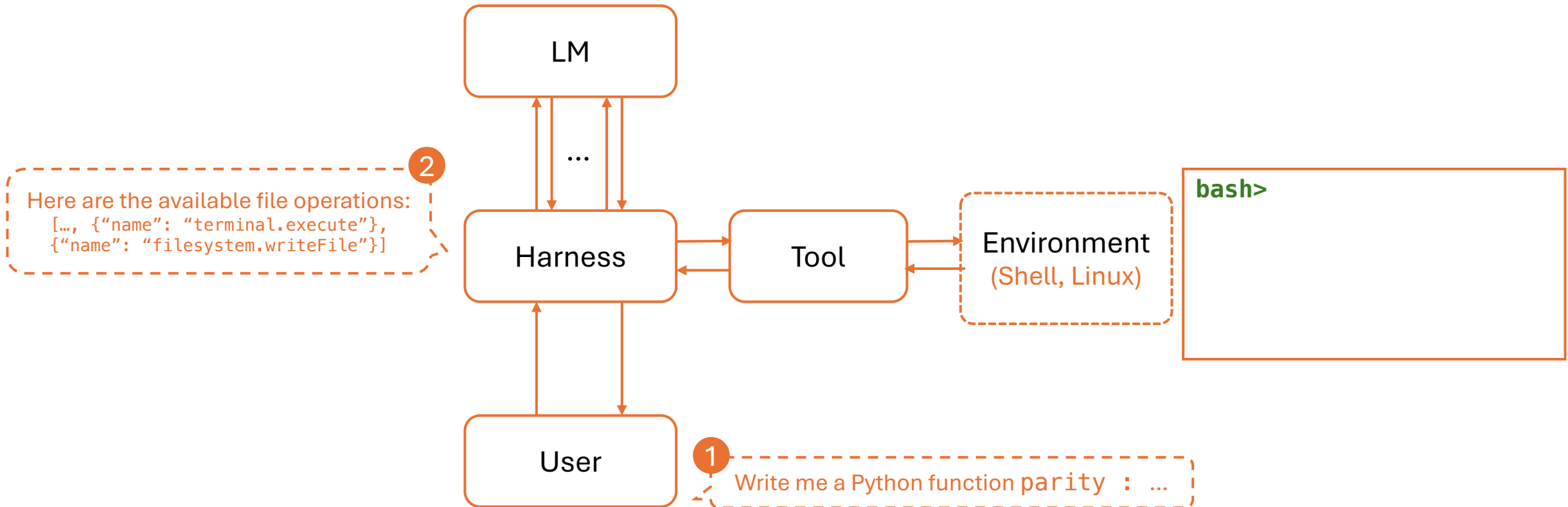
Terminal as Tool



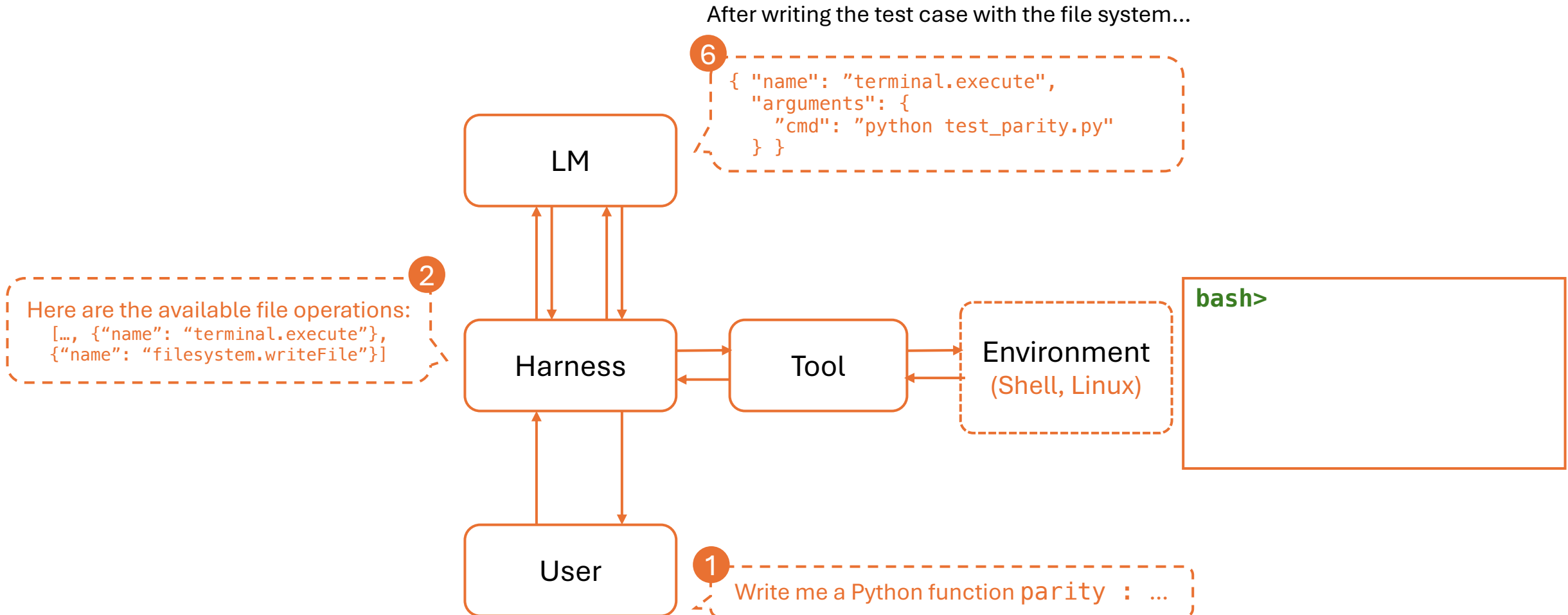
Terminal as Tool



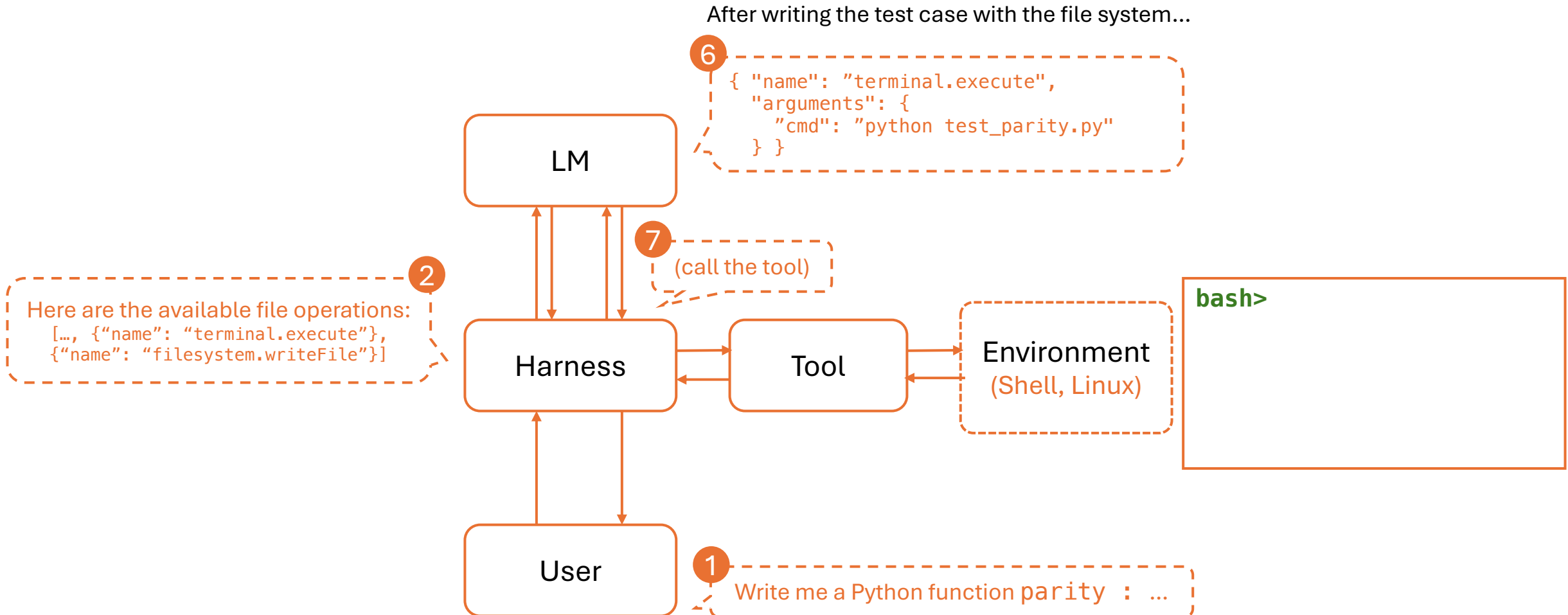
Terminal as Tool



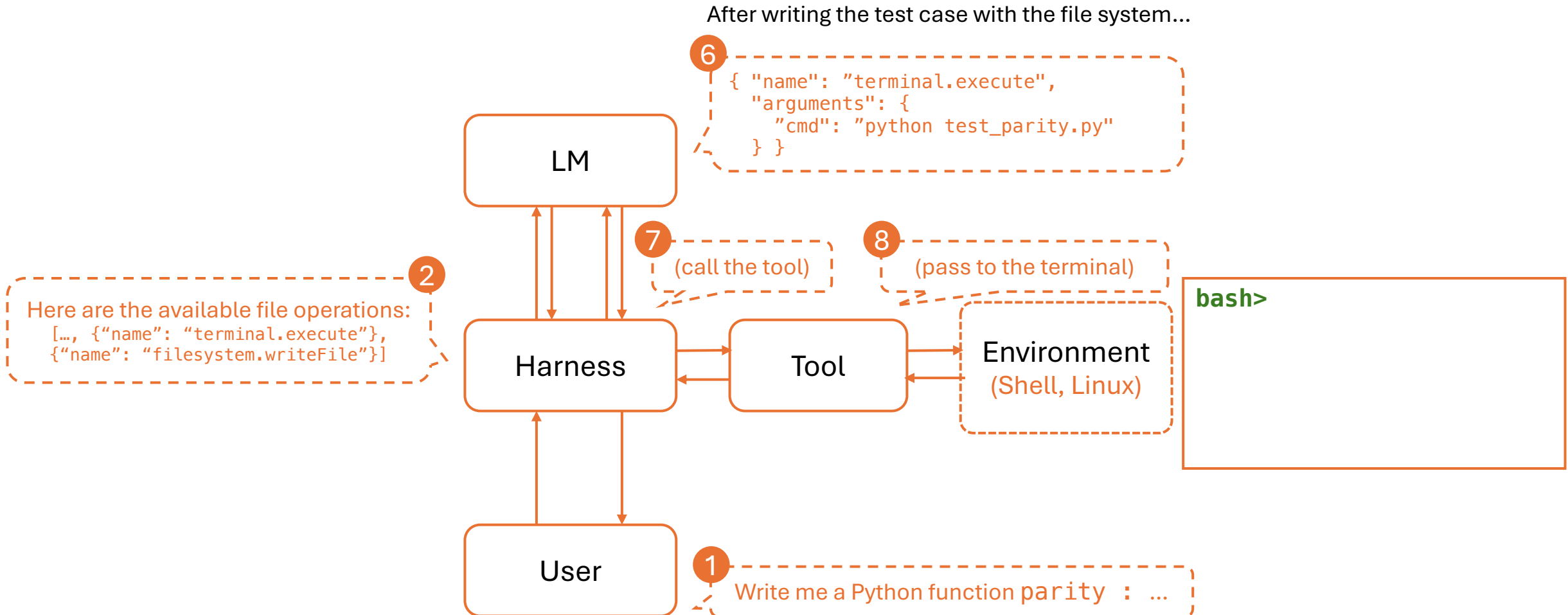
Terminal as Tool



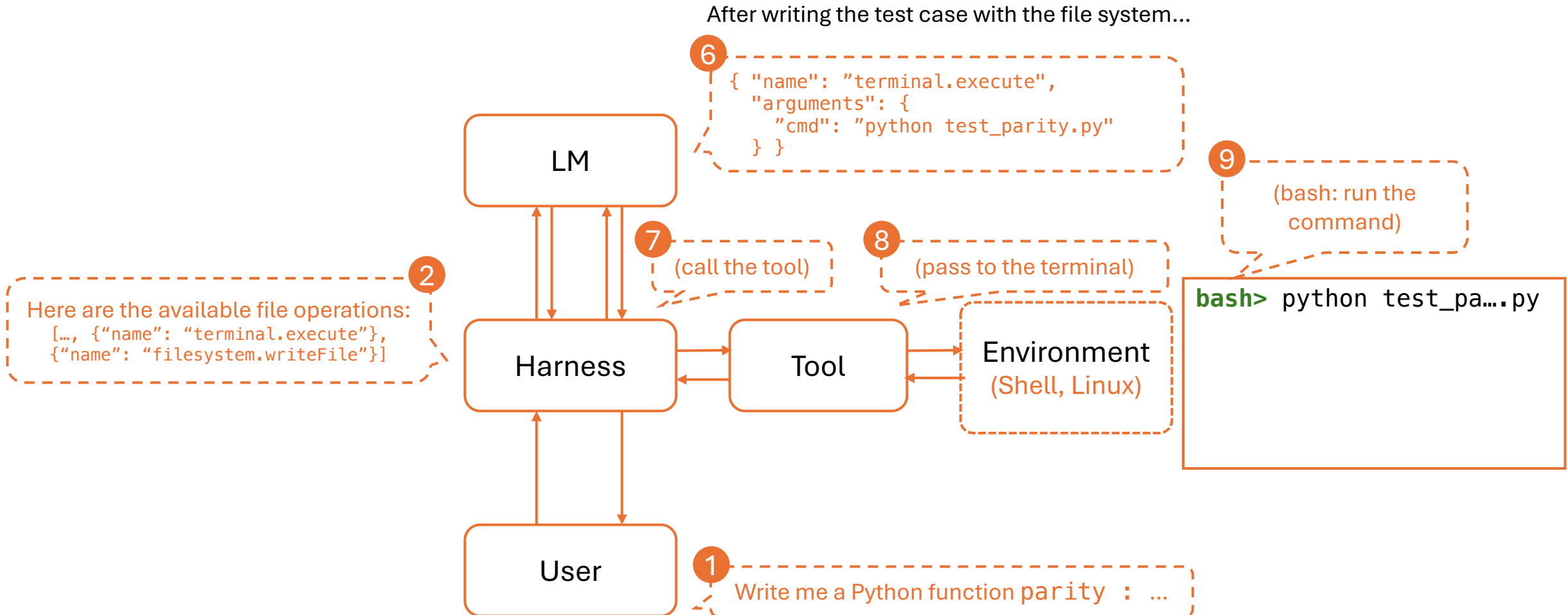
Terminal as Tool



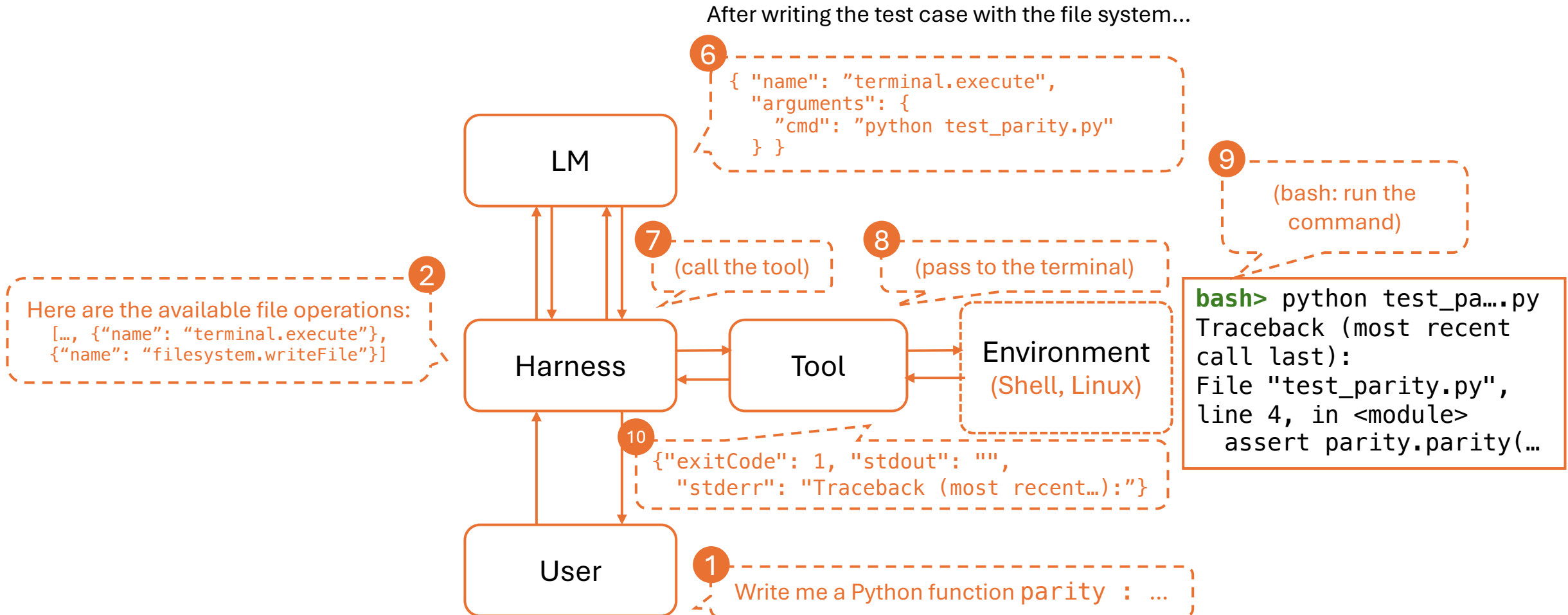
Terminal as Tool



Terminal as Tool



Terminal as Tool



Terminal as Tool

- **Context blow-up** from command output (esp. build logs)
 - Keep a separate “**bash context buffer**” from chat/planning memory.
 - Before sending to the LLM, budget-check size.
 - If too large: summarize head + tail (e.g., first 50 lines, last 200 lines, plus a bullet list of errors/warnings).
 - If still large: store full log in a vector DB (with run ID + command + timestamp); send only a compact summary + retrieval keys.
 - Always attach metadata: command, exit code, duration, bytes, truncation flag.

Can you execute a command for me brew install opam

@ 1 Tab

39.2%

Can you execute this command for me: top

Agent %I Auto



I'll execute the top command for you to show system processes and resource usage.

Rejected command: top

\$ top

[Terminal output truncated: ~3675KB dropped from beginning]

```
lightEx 0.0 00:00.03 3 2 43 2705K 0B 2384K 68189 1 sleeping *0[1] 0.
00000 0.00000 501 957 92 189 78 3805 384 209
3 0 0.0 0 0 40 liby99 N/A N/A N/A N/
A N/A N/A
68188 ContactsCoreSpot 0.0 00:00.05 3 2/1 47 3089K 0B 2736K 68188 1 sleeping
*0[1] 0.00000 0.00000 501 1445 103 249 126 4697 595
386 44 0 0.0 0 0 40 liby99 N/A N/
A N/A N/A N/A N/A
68074 sysmond 0.0 00:23.49 2 1 17 1809K 0B 1488K 68074 1 sleeping
0[2] 0.00000 0.00000 0 6922 80 377259 223020 448584 345814
92195 35 1 0.0 0 0 0 root N/A N/
A N/A N/A N/A N/A
67831 smd 0.0 00:00.21 2 1 30 2177K 0B 1888K 67831 1 sleeping
0[23] 0.00000 0.00000 0 4197 81 864 526 9237 2086
1873 124 2 0.0 0 0 0 root N/A N/
A N/A N/A N/A N/A
67822 countryd 0.0 00:00.28 2 1 34 3233K 0B 2832K 67822 1 sleeping
0[4] 0.00000 0.00000 0 2949 87 316 195 7464 646
1017 43 2 0.0 0 0 0 root N/A N/
A N/A N/A N/A N/A
67029 iCloudNotificati 0.0 00:00.76 3 1 101 6705K 0B 5072K 67029 1 sleeping
```

Use Allowlist

Rejected

Terminal as Tool

- Don't rely on stdout alone, **inspect exit code & stderr**
 - Treat **exitCode** as the primary **success signal**.
 - Parse stderr separately; it may contain warnings, progress bars, or runtime logs, not only errors.
 - Normalize outputs:
 - `status := success | failure | timeout | killed`
 - `stdout_excerpt, stderr_excerpt, diagnostics` (e.g., grep for “error:”, “warning:”).
- Prefer **structured extraction** (regex for **file:line:col**, error codes) to feed precise hints back to the LLM.

Terminal as Tool

- **Non-terminating** / long-running commands
 - Set **hard timeouts** per command (e.g., 30–120s default; shorter for cat, longer for pytest).
 - For allowed long runs:
 - Stream **incremental chunks** (e.g., every N seconds / N KB) and ask the LLM: “Continue or preempt?”
 - Support **preemption** (SIGINT/SIGKILL), and return a partial transcript with a “truncated” marker.
 - Maintain a deny/guard list (e.g., top, interactive shells, tail -f) unless explicitly whitelisted.

OPENHANDS: AN OPEN PLATFORM FOR AI SOFTWARE DEVELOPERS AS GENERALIST AGENTS

**Xingyao Wang^{1,10}, Boxuan Li², Yufan Song², Frank F. Xu², Xiangru Tang³,
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Hoang H. Tran⁸, Fuqiang Li, Ren Ma, Mingzhang Zheng, Bill Qian³, Yanjun Shao³,
Niklas Muennighoff⁵, Yizhe Zhang, Binyuan Hui⁹, Junyang Lin⁹,
Robert Brennan¹⁰, Hao Peng¹, Heng Ji¹, Graham Neubig^{2,10}**

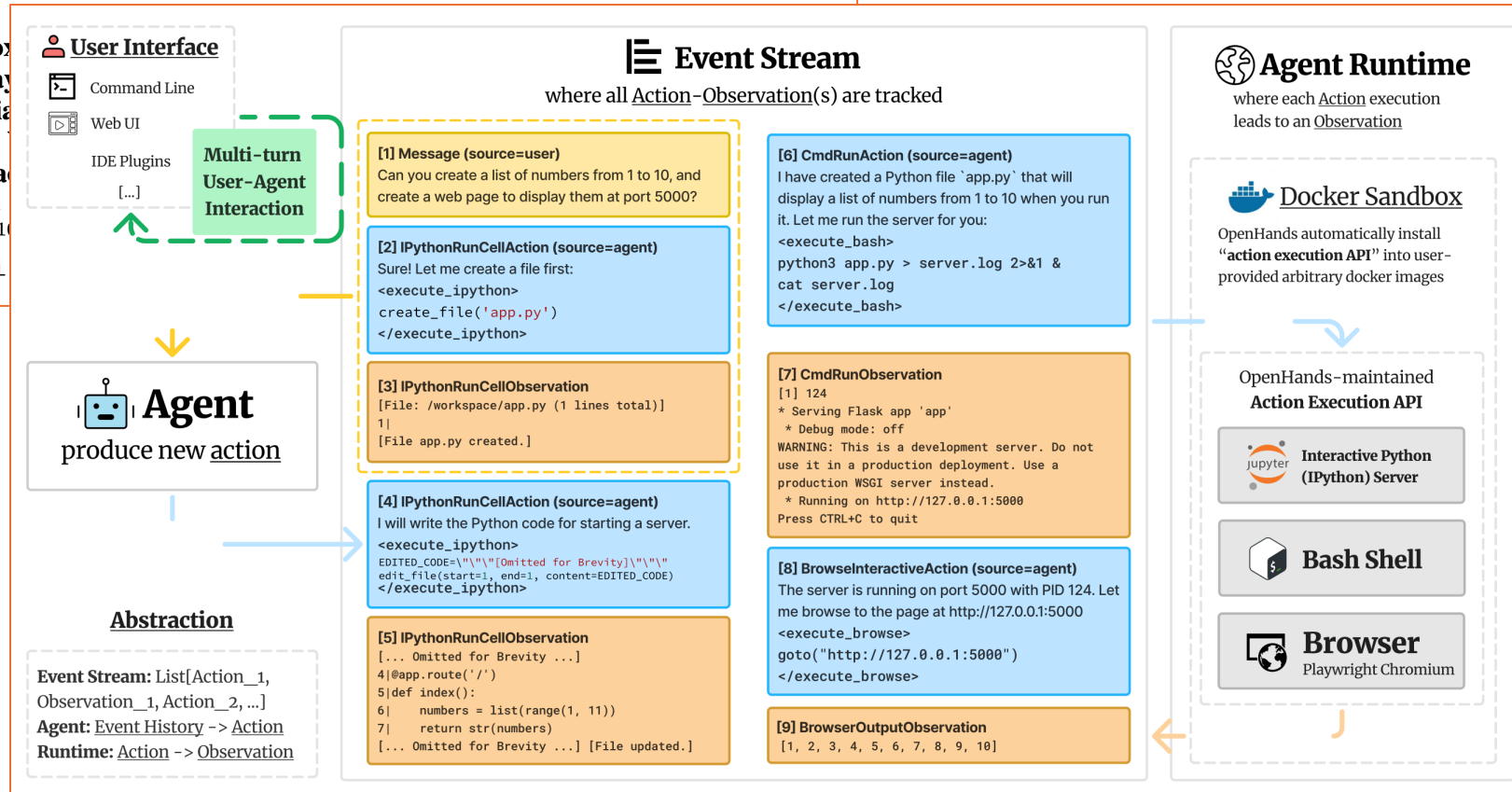
¹UIUC ²CMU ³Yale ⁴UC Berkeley ⁵Contextual AI ⁶KAUST ⁷ANU

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👋 OPENHANDS: AN OPEN PLATFORM FOR AI SOFTWARE DEVELOPERS AS GENERALIST AGENTS

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OPENHANDS: AN OPEN PLATFORM FOR

AI SOFTWARE DEVELOPERS AS GENERALIST AGENTS

 **User Interface**
 Command Line
 Web UI
 IDE Plugins
 Multi-turn User-Agent Interaction

 **Event Stream**
 where all Action-Observation(s) are tracked

 **Agent Runtime**
 where each Action execution leads to an Observation

 **Agent**
 produce new action

Abstraction
 Event Stream: List(Action_1, Observation_1, Action_2, ...)
 Agent: Event History -> Action
 Runtime: Action -> Observation

Table 2: Evaluation benchmarks in OpenHands.

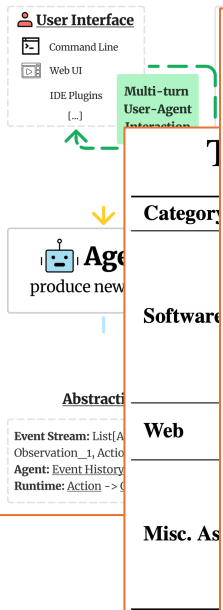
Category	Benchmark	Required Capability
Software	SWE-Bench (Jimenez et al., 2024)	Fixing Github issues
	HumanEvalFix (Muennighoff et al., 2024)	Fixing Bugs
	BIRD (Li et al., 2023b)	Text-to-SQL
	BioCoder (Tang et al., 2024c)	Bioinformatics coding
	ML-Bench (Tang et al., 2024b)	Machine learning coding
	Gorilla APIBench (Patil et al., 2023)	Software API calling
	ToolQA (Zhuang et al., 2024)	Tool use
Web	WebArena (Zhou et al., 2023a)	Goal planning & realistic browsing
	MiniWoB++ (Liu et al., 2018)	Short trajectory on synthetic web
Misc. Assistance	GAIA (Mialon et al., 2023)	Tool-use, browsing, multi-modality
	GPQA (Rein et al., 2023)	Graduate-level Google-proof Q&A
	AgentBench (Liu et al., 2023)	Operating system interaction (bash)
	MINT (Wang et al., 2024b)	Multi-turn math and code problems
	Entity Deduction Arena (Zhang et al., 2024a)	State tracking & strategic planning
	ProofWriter (Tafjord et al., 2021)	Deductive Logic Reasoning



OPENHANDS: AN OPEN PLATFORM FOR

AI S

Xing
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Hoar
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Robe
1UIU
8HCH
xing



OpenHands: Code Less, Make More

CONTRIBUTORS

378

STARS

64K

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SLACK

JOIN US



PROJECT

CREDITS



DOCUMENTATION



PAPER ON ARXIV

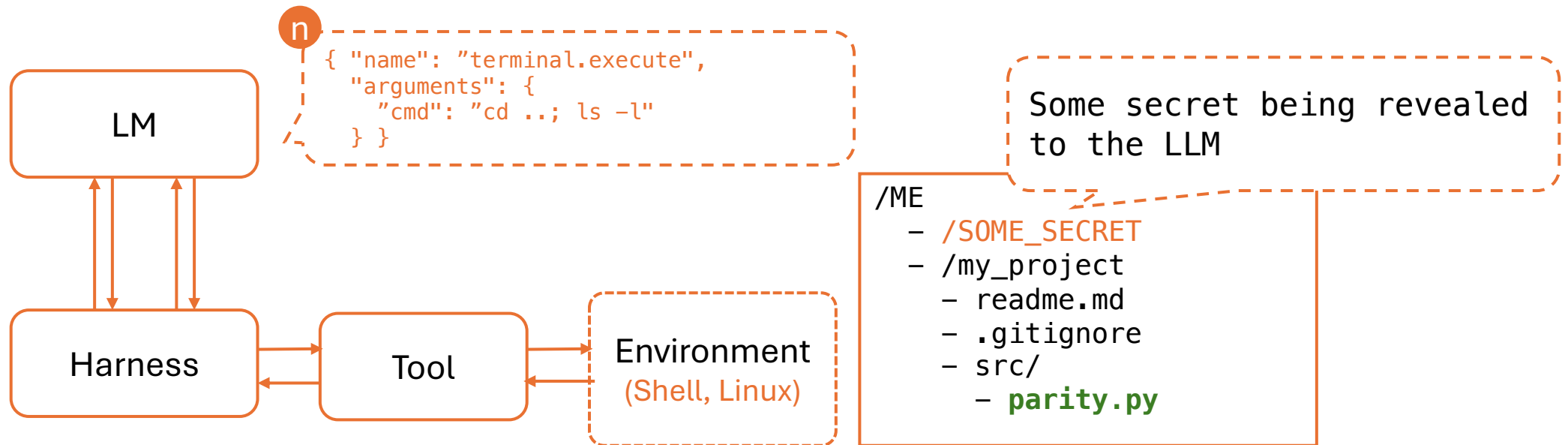


BENCHMARK SCORE

[Deutsch](#) | [Español](#) | [français](#) | [日本語](#) | [한국어](#) | [Português](#) | [Русский](#) | [中文](#)

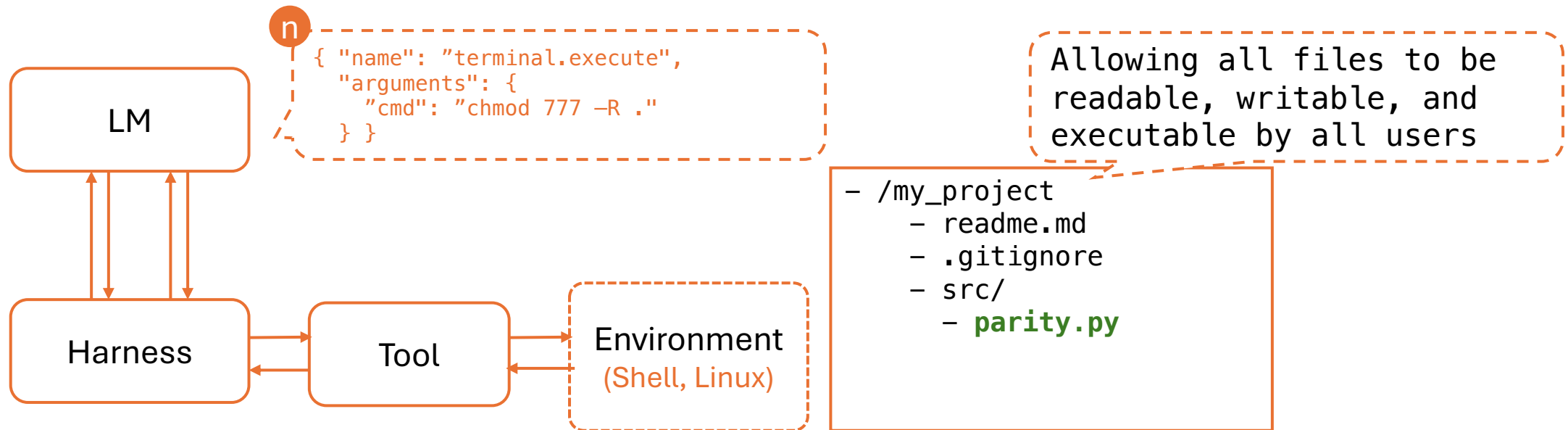
Terminal as Tool: Security

- LLM may leave the current working directory
 - CWE-22: Path Traversal Vulnerability



Terminal as Tool: Security

- LLM may pursue excessive permissions
 - CWE-284: Improper Access Controls



When Developer Aid Becomes Security Debt: A Systematic Analysis of Insecure Behaviors in LLM Coding Agents

Matous Kozak

Microsoft

Czech Technical University in Prague

Roshanak Zilouchian Moghaddam

Microsoft

Siva Sivaraman

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When Developer Aid Becomes Security Debt: A Systematic Analysis of Insecure Behaviors in LLM Coding Agents

M

Czech Techni

2.1 Data Collection

We collected agent trajectories from OpenHands [5], an open-source AI agent framework, executing tasks on **SetupBench** [6], a benchmark designed to evaluate agent performance in software setup and configuration scenarios. *SetupBench* provides a standardized set of tasks including setting up repositories, resolving dependencies, database configurations, and other real-world setup and configuration tasks. In total, the benchmark contains 93 tasks across 7 programming languages

(Python, TypeScript, JavaScript, Java, Go, Rust, C++). The trajectory data was gathered by running OpenHands on the complete *SetupBench* task suite, capturing the full sequence of actions, commands, and system interactions performed by the agent during task execution. We ran the agent in a controlled Linux environment (Ubuntu 22.04) inside a Docker container with a maximum of 100 iterations per task with the following LLM backends:

When Developer Aid Becomes Security Debt: A Systematic Analysis of Insecure Behaviors in LLM Coding Agents

2.1 Data Collection

We collected agent trajectories from OpenHands [5], an open-source AI agent framework, executing tasks on `SecWiki` and configuring repositories and system configurations.

Czech

(Python, TypeScript, OpenHands, and system configurations) Linux environment task with the

Table 1: Taxonomy of insecure behaviors observed in agent trajectories.

CWE Category	Description	Example
CWE-200: Exposure of Sensitive Information ¹	Storing credentials or secrets directly in code or scripts rather than secure storage	Embedding database credentials in script files or passing passwords via command-line
CWE-284: Improper Access Control ²	File or directory permissions that are overly broad, violating least privilege	Using <code>chmod 777</code> which grants all users full control over a file
CWE-494: Download of Code Without Integrity Check ³	Running scripts from external sources without verification	Piping <code>curl</code> commands to shell (“ <code>curl-bash</code> ”) which can execute untrusted code
CWE-693: Protection Mechanism Failure ⁴	Intentionally disabling security mechanisms to avoid obstacles	Skipping SSH host key checks or disabling database authentication

When Developer Aid Becomes Security Debt: A Systematic Analysis of Insecure Behaviors in LLM Coding Agents

2.1 Data Collection

We collected agent trajectories from the *SetupBench* tasks on *Set* and configuration tasks on *SetupBench*.

Table 1: Taxonomy of insecure behaviors observed in agent trajectories.

CWE Category	Description	Example
CWE-200: Buffer Overflow	Agent writes to a buffer without checking its bounds.	<code>strcpy(dest, src);</code> where <code>dest</code> is not null-terminated.

Table 2: Distribution of agent’s insecure steps and trajectories from the *SetupBench* by model.

	Steps			Trajectories		
	Total	Insecure	[%]	Total	Insecure	[%]
GPT-4o	1784	34	1.91	93	15	16.13
GPT-4.1	2342	21	0.90	92	16	17.39
Claude 3.5 Sonnet	1236	38	3.07	85	17	20.00
Claude 3.7 Sonnet	3185	62	1.95	92	21	22.83
Claude 4 Sonnet	3915	73	1.86	93	25	26.88
Average			1.83			20.66

Leaderboards

BENCHMARKS

SWE-bench

SWE-bench Verified

SWE-bench Multilingual

SWE-bench Multimodal

SWE-bench Lite

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Official Leaderboards

[Verified](#)[Multimodal](#)[Multilingual](#)[Lite](#)[Full](#)

Verified is a human-filtered subset of 500 instances ([details](#)). Defaults to [bash-only](#) setting (run with [mini-SWE-agent](#)).

[📊 Compare results](#)☒ Bash Only

AGENT mini-SWE-agent ×

☐	#	MODEL	AGENT	% RESOLVED ▼	AVG. \$	TRAJS	ORG	DATE	RELEASE
☐	1	Claude 4.5 Opus (high)	mini-SWE-agent	76.80	\$0.75	🔗	AI	2026-02-17	2.0.0
☐	2	Gemini 3 Flash (high)	mini-SWE-agent	75.80	\$0.36	🔗	🌟	2026-02-17	2.0.0
☐	3	MiniMax M2.5 (high)	mini-SWE-agent	75.80	\$0.07	🔗		2026-02-17	2.0.0
☐	4	Claude 4.6 Opus	mini-SWE-agent	75.60	\$0.55	🔗	AI	2026-02-17	2.0.0
☐	5	Claude 4.5 Opus (medium)	mini-SWE-agent	74.40	\$0.72	🔗	AI	2025-11-24	1.16.0
☐	6	Gemini 3 Pro Preview	mini-SWE-agent	74.20	\$0.46	🔗	🌟	2025-11-18	1.15.0
☐	7	GLM 5 (high)	mini-SWE-agent	72.80	\$0.53	🔗		2026-02-17	2.0.0
☐	8	GPT 5.2 (high)	mini-SWE-agent	72.80	\$0.47	🔗		2026-02-17	2.0.0



Leaderboards

BENCHMARKS

SWE-bench
SWE-bench Verified
SWE-bench Multilingual
SWE-bench Multimodal
SWE-bench Lite

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Bash Only: Comparing Language Models

While the full leaderboard compares arbitrary systems, we are also interested in evaluating language models directly. To make an apples-to-apples comparison of LMs easier, we evaluate all LMs using [mini-SWE-agent](#) in a minimal bash environment. No tools, no special scaffold structure; just a simple [ReAct](#) agent loop. These results represent the state-of-the-art LM performance when given just a bash shell and a problem.

On the leaderboard, use the **Agent** dropdown to select between the mini-SWE-agent results and the full leaderboard with all agents.

▼ Click for more details on the bash-only setup

- We use [this configuration](#) for all models.
- The release number in the leaderboard corresponds to the version of the mini-SWE-agent used to run the evaluation.
- Results of release 1.x and 2.x are not necessarily comparable to each other, as 2.x uses tool calling to invoke actions, whereas 1.x parses action from the output strings. Read more about the changes in the [mini-SWE-agent v2 migration guide](#).
- For all results of release 1.x and earlier, the LM temperature is set to 0.0 if the temperature parameter is supported. For all results of release 2.x and later, the temperature parameter is not set.
- Other than the aforementioned notes, small changes in the setup and configuration are captured by the version number in the leaderboard. Version numbers correspond to tags in the mini-SWE-agent repository. Since the mini-SWE-agent repository contains other components as well, a new version number does not necessarily mean that anything of relevance has changed for the bash-only leaderboard setting. We do *not* aim to tune the configuration and setup to reach higher and higher scores. Instead, we only make general fixes to the framework, as well as clarifications in the prompt to provide a maximally fair evaluation setup for the LMs. Generally, everything in the minor or patch release version number should be a minor change for the purpose of the bash-only leaderboard.
- [This guide](#) shows how to run the evaluation yourself.

Bash Only: Comparing Language Models

 **SWE-bench**

Leaderboards

BENCHMARKS

SWE-bench

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 **Official Leaderboards**

Verified

Multimodal

Multilingual

Lite

Full

Verified is a human-filtered subset of 500 instances (details). Defaults to *bash-only* setting (run with *mini-SWE-agent*).

Compare results

Bash Only

AGENT mini-SWE-agent

	#	MODEL	AGENT	% RESOLVED	AVG. \$	TRAJS	ORG	DATE	RELEASE
☐	1	Claude 4.5 Opus (high)	mini-SWE-agent	76.80	\$0.75			2026-02-17	2.0.0
☐	2	Gemini 3 Flash (high)	mini-SWE-agent	75.80	\$0.36			2026-02-17	2.0.0
☐	3	MiniMax M2.5 (high)	mini-SWE-agent	75.80	\$0.07			2026-02-17	2.0.0

Compare results

Bash Only

	#	MODEL	AGENT	% RESOLVED	ORG	DATE	SITE
☐	1	Claude 4.5 Opus	Sonar Foundation Agent	79.20		2025-12-05	
☐	2	Claude 4.5 Opus (medium)	live-SWE-agent	79.20		2025-12-15	
☐	3	Doubao-Seed-Code	TRAE	78.80		2025-09-28	
☐	4	Gemini 3 Pro Preview	live-SWE-agent	77.40		2025-11-20	
☐	5	Claude 4 Sonnet	EPAM AI/Run Develop...	76.80		2025-08-04	
☐	6	Multiple	Atlassian Rovo Dev	76.80		2025-09-02	
☐	7	Claude 4.5 Opus (high)	mini-SWE-agent	76.80		2026-02-17	
☐	8	Multiple	ACoder	76.40		2025-08-19	

Bash Only: Comparing Language Models

SWE-bench

Leaderboards

BENCHMARKS

- SWE-bench
- SWE-bench Verified
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- SWE-bench Lite

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Official Leaderboards

Verified Multimodal Multilingual Lite Full

Verified is a human-filtered subset of 500 instances (details). Defaults to bash-only setting (run with mini-SWE-agent).

Compare results Bash Only AGENT mini-SWE-agent x

☐	#	MODEL	AGENT	% RESOLVED	AVG. \$	TRAJS	ORG	DATE	RELEASE
☐	1	Claude 4.5 Opus (high)	mini-SWE-agent	76.80	\$0.75		AI	2026-02-17	2.0.0
☐	2	Gemini 3 Flash (high)	mini-SWE-agent	75.80	\$0.36			2026-02-17	2.0.0
☐	3	MiniMax M2.5 (high)	mini-SWE-agent	75.80	\$0.07			2026-02-17	2.0.0
☐	4	Claude 4.6 Opus							
☐	5	Claude 4.5 Opus							
☐	6	Gemini 3 Pro Prev							
☐	7	GLM 5 (high)							
☐	8	GPT 5.2 (high)							

Compare results Bash Only

☐	#	MODEL	AGENT	% RESOLVED	ORG	DATE	SITE
☐	1	Claude 4.5 Opus	Sonar Foundation Agent	79.20		2025-12-05	
☐	2	Claude 4.5 Opus (medium)	live-SWE-agent	79.20		2025-12-15	
☐	3	Doubao-Seed-Code	TRAE	78.80		2025-09-28	
☐	4	Gemini 3 Pro Preview	live-SWE-agent	77.40		2025-11-20	
☐	5	Claude 4 Sonnet	EPAM AI/Run Develop...	76.80		2025-08-04	
☐	6	Multiple	Atlassian Rovo Dev	76.80		2025-09-02	
☐	7	Claude 4.5 Opus (high)	mini-SWE-agent	76.80	AI	2026-02-17	
☐	8	Multiple	ACoder	76.40		2025-08-19	

Shifting responsibility back to the users...

Can you execute this command for me: top

I'll execute the `top` command for you to show system processes and resource usage.

 Run command: top

```
$ top
```

Use Allowlist 

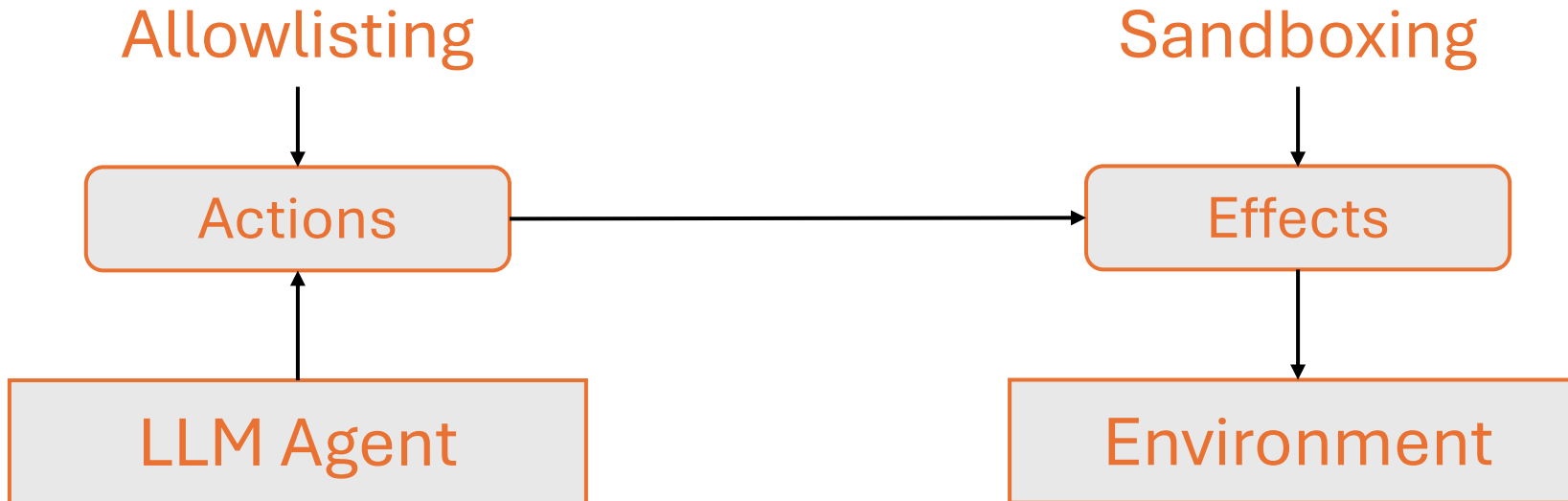
Skip

Add 'top' to Allowlist  

Run 

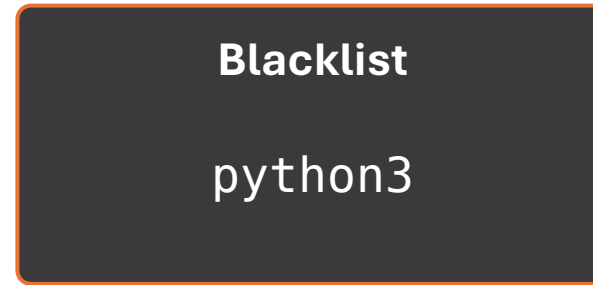
Terminal Access: What are we defending against?

- The terminal is the widest attack surface in an agentic framework
- How to defend? **Allowlisting** + **Sandboxing**



Hypothetical Situation

Allowlisting



(only checks the first token of the bash command)

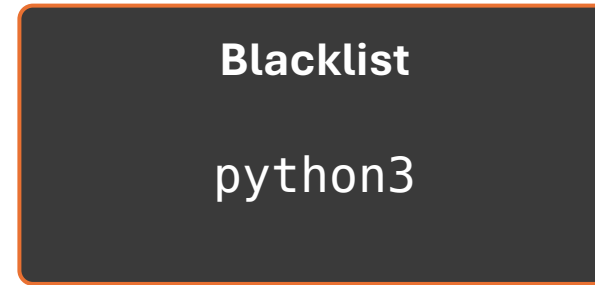
LLM-Agent Proposed Command:

cd foo/



Hypothetical Situation

Allowlisting



(only checks the first token of the bash command)

LLM-Agent Proposed Command:

cd foo/

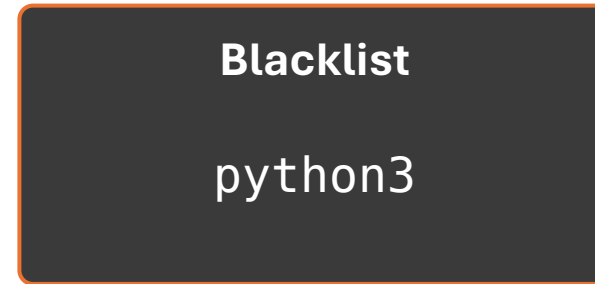


python3 test.py



Hypothetical Situation

Allowlisting



(only checks the first token of the bash command)

LLM-Agent Proposed Command:

cd foo/



python3 test.py

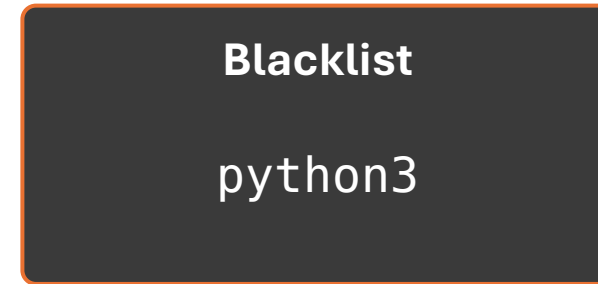


cd foo/.. ; python3 test.py



Hypothetical Situation

Allowlisting



(only checks the first token of the bash command)

LLM-Agent Proposed Command:

cd foo/



python3 test.py



cd foo/.. ; python3 test.py



← This is bad

Even after we fix the parsing, it could still leak

- Interpreters are universal — `python3 -c "import os; os.system(...)"`
- Indirection hides the payload — `bash -c, eval, xargs, make`
- Substitution — `$(...), backticks, <(...)`
- Encoding — `echo ... | base64 -d | sh`
- Paths are not syntax — `..., symlinks, globs, $HOME`

High Level Picture

